

Airline Delay and Operations Intelligence

Houston, 2011 — 227,496 flights

DSA 3050A — Business Intelligence & Visualization · Group 05

J. P. Chawanda · Catherine Ingabire · Ruth Musanhu · Faith Chakwanira · Racheal H. Skiven-data

Part 1: Airline Delay and Operations Intelligence

The Business Problem

Houston Airport System — Operations Performance

- Delay is the largest controllable cost in airport operations — fuel, crew duty limits, gate reallocation, missed connections, compensation.
- Houston ran 227,496 departures in 2011. 47.8% of operated flights arrived late.
- Management could see that the network was late — not where the lateness was being created.
- Without attribution, mitigation spending is guesswork.

Our brief: attribute delay and cancellation to the carrier, route, airport, hour and month that produced them.

The Data

hflights — Houston departures, 2011

227,496

FLIGHTS

21

COLUMNS

15

AIRLINES

149

ROUTES

116

DESTINATIONS

- Source: US DOT Bureau of Transportation Statistics, via the R *hflights* package — genuine public government reporting.
- Period: every commercial departure from IAH and HOU in 2011.
- Structurally clean: zero duplicates, zero blank rows — but times stored as HHMM integers, flags as 0/1, and a 98.69% null rate on cancellation code.

Cleaning and Modelling

One fact table, three dimensions, 22 measures

- All 10 required cleaning tasks, and 8 advanced tasks against a requirement of 6.
- Renamed 13 columns to business language; decoded every code and flag.
- Split HHMM integers into real clock times; merged origin and destination into a Route key.
- Derived Flight Status, Delay Category, Distance Band, Time of Day, Season and Delay Recovered.
- Star schema: Fact_Flights + Dim_Date + Dim_Airline + a Group By aggregate table, measures isolated in _Measures.

The decision that mattered: 3,622 null arrival delays belong to cancelled and diverted flights. We left them null. Filling them with zero would have counted 3,622 flights that never arrived as "on time".

Part 2: Power Query Data Preparation and Transformation

1. Importing the Data

The screenshot shows a data import interface. On the left, a 'Navigator' pane displays a search bar, 'Display Options', and a list of data sources: 'H flight dataset.xlsx [2]', 'Table1', and 'hflights'. The 'hflights' source is selected with a checkmark. On the right, a preview of the 'hflights' table is shown with columns: Year, Month, DayofMonth, DayOfWeek, DepTime, ArrTime, and Unique. The table contains 23 rows of data for the year 2011, month 1. At the bottom, there are three buttons: 'Load' (highlighted in green), 'Transform Data', and 'Cancel'.

Year	Month	DayofMonth	DayOfWeek	DepTime	ArrTime	Unique
2011	1	1	6	1400	1500	AA
2011	1	2	7	1401	1501	AA
2011	1	3	1	1352	1502	AA
2011	1	4	2	1403	1513	AA
2011	1	5	3	1405	1507	AA
2011	1	6	4	1359	1503	AA
2011	1	7	5	1359	1509	AA
2011	1	8	6	1355	1454	AA
2011	1	9	7	1443	1554	AA
2011	1	10	1	1443	1553	AA
2011	1	11	2	1429	1539	AA
2011	1	12	3	1419	1515	AA
2011	1	13	4	1358	1501	AA
2011	1	14	5	1357	1504	AA
2011	1	15	6	1359	1459	AA
2011	1	16	7	1359	1509	AA
2011	1	17	1	1530	1634	AA
2011	1	18	2	1408	1508	AA
2011	1	19	3	1356	1503	AA
2011	1	20	4	1507	1622	AA
2011	1	21	5	1357	1459	AA
2011	1	22	6	1355	1456	AA
2011	1	23	7	1356	1501	AA

The hflights table is selected, then sent to Transform Data instead of Load.

1. Data in the power query

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Query Settings

Layout Data Preview Columns Parameters Advanced Dependencies

Queries [2]

- hflights
- hflights (2)

	1 ² ₃ Year	1 ² ₃ Month	1 ² ₃ DayofMonth	1 ² ₃ DayOfWeek	ABC 123 DepTime
1	2011	1	1	6	
2	2011	1	2	7	
3	2011	1	3	1	
4	2011	1	4	2	
5	2011	1	5	3	
6	2011	1	6	4	
7	2011	1	7	5	
8	2011	1	8	6	
9	2011	1	9	7	
10	2011	1	10	1	

Query Settings

PROPERTIES

Name

hflights (2)

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type

Power Query Editor after load. Two queries exist (hflights, hflights (2)). Auto steps: Source, Navigation, Promoted Headers, Changed Type.

2. Basic Cleaning

2.1 Rename and Delete Columns

	ABC 123	Tax out Time	1 ² ₃ Cancelled	1 ² ₃ Diverted
7		13	0	0
6		9	0	0
5		17	0	0
9		22	0	0
9		9	0	0
6		13	0	0
12		15	0	0
7		12	0	0
8		22	0	0
6		19	0	0
8		20	0	0
4		11	0	0

Query Settings

PROPERTIES

Name

hflights (2)

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

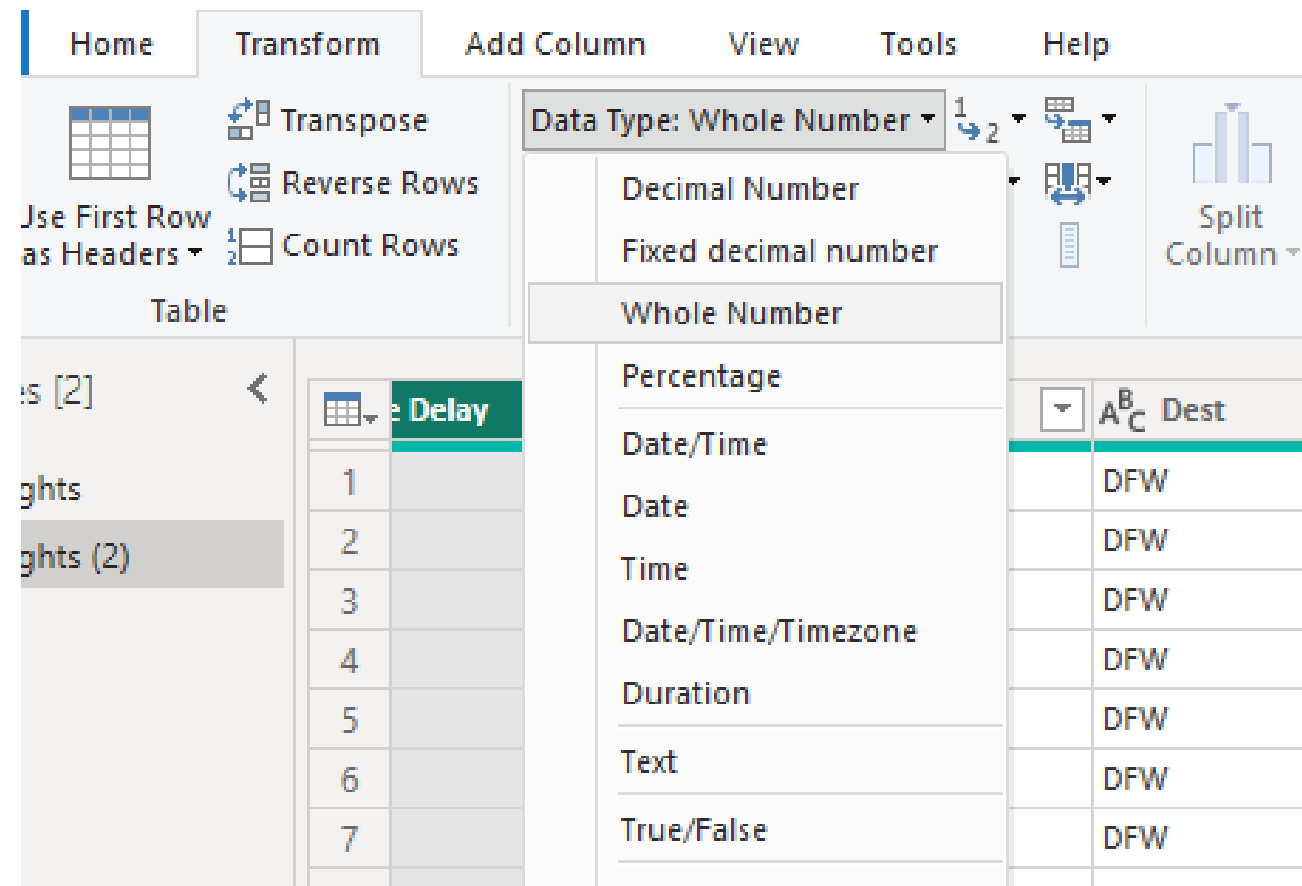
Changed Type

Renamed Columns

X Removed Columns

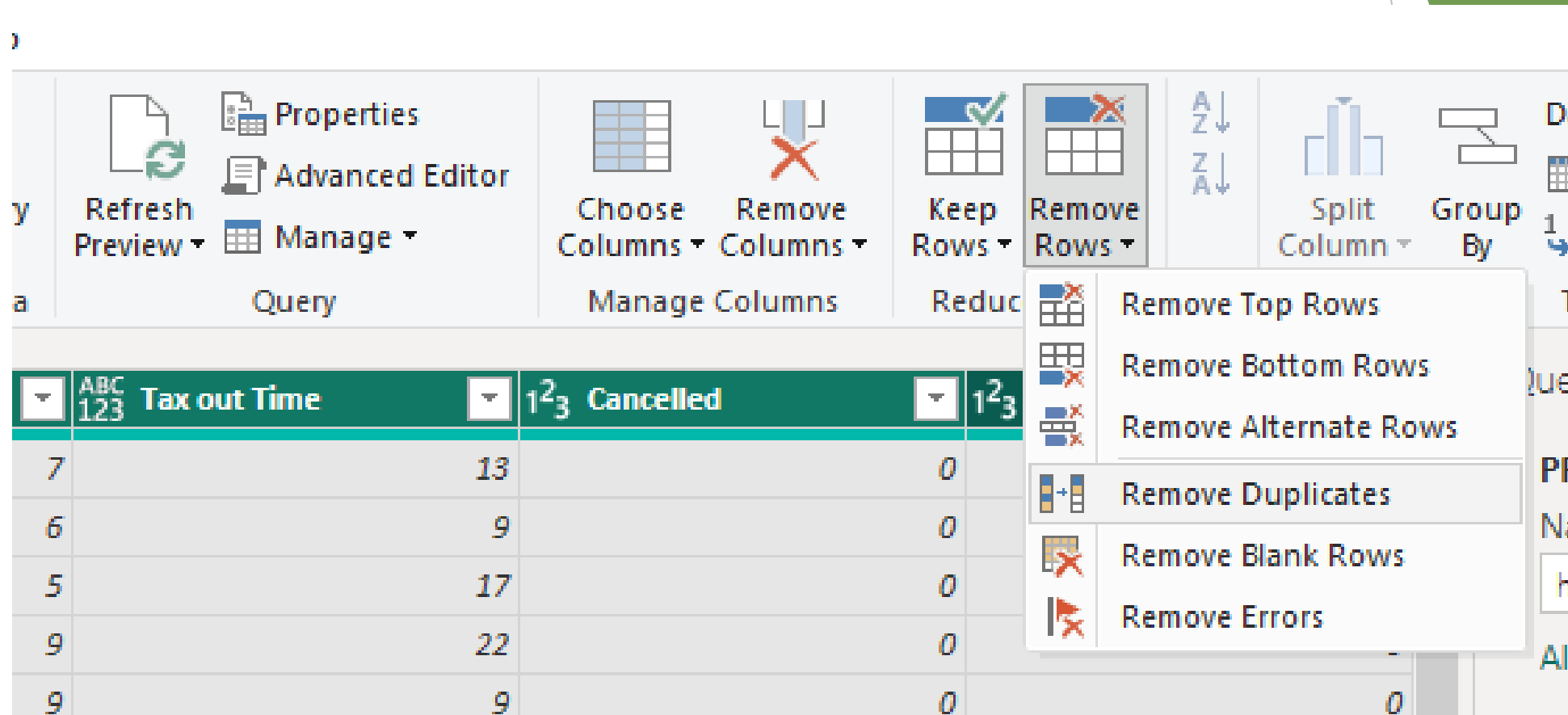
Columns renamed, then unneeded columns (Cancelled, Diverted) removed.

2.2 Change the Data Types



Data type set to Whole Number for a numeric column.

2.3 Remove Duplicate and Blank Rows



Remove Rows > Remove Duplicates, then Remove Blank Rows.

2.4 Trim Text and Clean

Format > Trim applied to remove extra spaces from text columns.

IAH	DFW	Renamed Columns
IAH	DFW	Removed Columns
IAH	DFW	Changed Type1
IAH	DFW	Removed Duplicates
IAH	DFW	Removed Blank Rows
IAH	DFW	Trimmed Text
IAH	DFW	✕ Cleaned Text

2.5 Replace Inconsistent Values

12

ABC Cancelled

123 Diverted

13	No	
9	No	
17	No	
22	No	
9	No	
13	No	
15	No	
12	No	
22	No	

Query Settings

PROPERTIES

Name

hflights (2)

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Replace Values

Replace one value with another in the selected columns.

Value To Find

1

Replace With

Yes

Advanced options

OK

Cancel

Replace Values coded entries (e.g. "1") replaced with readable labels (e.g. "Yes").

2.6 Handle Missing Values

"NA" entries left blank/null instead of filled in, so averages aren't skewed.

2.7 Remove Unnecessary Columns

	1 ² ₃ Tax in Time	1 ² ₃ Tax out
24	7	
24	6	
24	5	
24	9	
24	9	
24	6	
24	12	
24	7	
24	8	
24	6	

 Copy

 Remove Columns

Remove Other Columns

 Add Column From Examples...

Remove Duplicates

Remove Errors

 Replace Values...

Fill

Change Type

Transform

Right-click > Remove Columns to drop further unneeded fields (e.g. taxi times).

3.1 Fix Flight Date Type

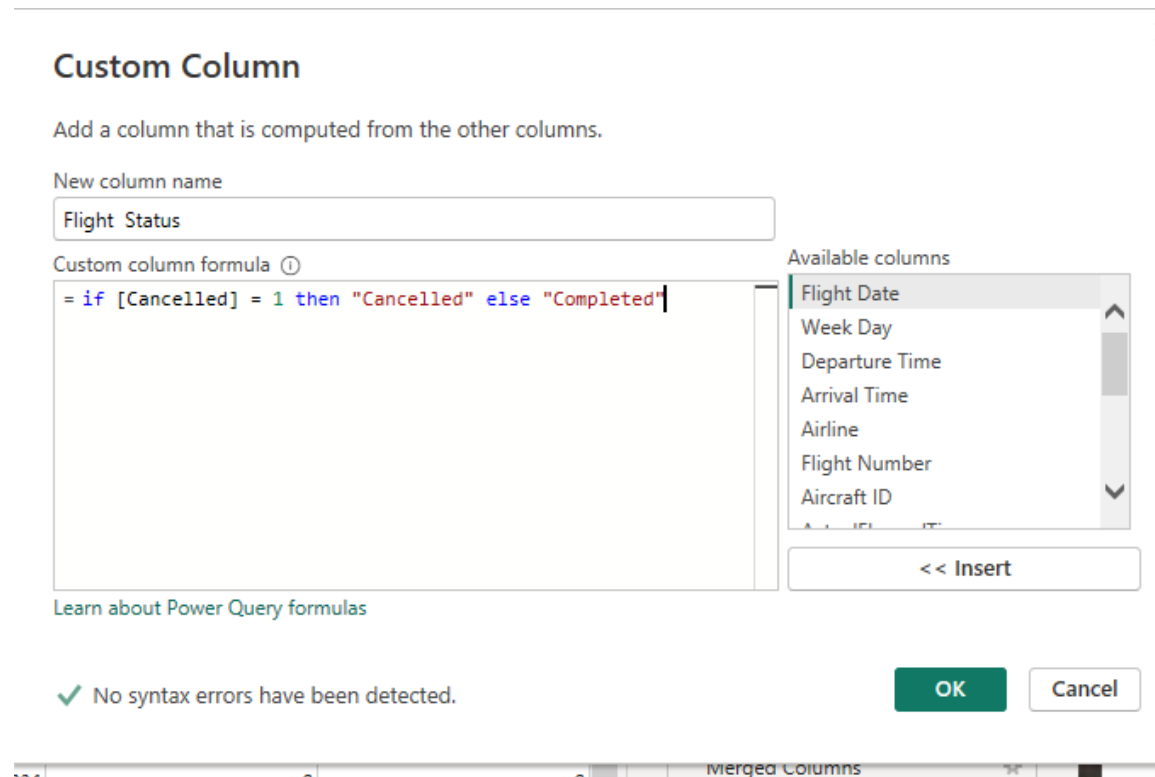
Flight	Percentage	Day
	Date/Time	
	Date	
	Time	
	Date/Time/Timezone	
	Duration	
	Text	
	True/False	
	Binary	
	09/01/2011	
	10/01/2011	
	11/01/2011	
	12/01/2011	
	13/01/2011	
	14/01/2011	
	15/01/2011	
	16/01/2011	
	17/01/2011	
	18/01/2011	
	19/01/2011	

Flight Date column type set to Date for correct sorting and filtering.

3.2 Create a Custom Column

Custom column "Flight Status" added:

```
= if [Cancelled] = 1 then "Cancelled" else "Completed"
```



Custom Column

Add a column that is computed from the other columns.

New column name
Flight Status

Custom column formula ⓘ
= if [Cancelled] = 1 then "Cancelled" else "Completed"

Available columns

- Flight Date
- Week Day
- Departure Time
- Arrival Time
- Airline
- Flight Number
- Aircraft ID

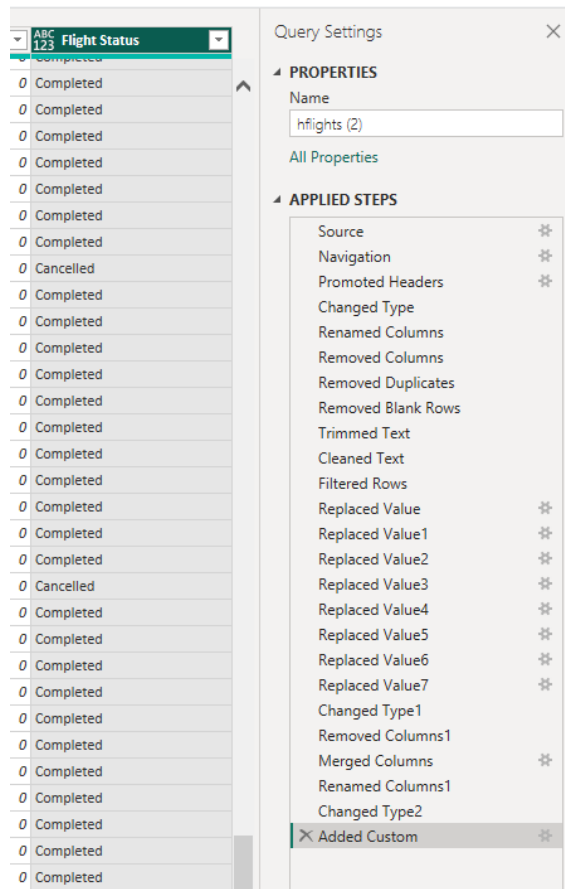
<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK Cancel

3.2 Create a Custom Column (cont.)



Flight Status column populated; Applied Steps show the full pipeline so far.

3.3 Create a Conditional Column

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Delay Category

	Column Name	Operator	Value ①		Output ①
If	Arrival Delay	is less than or equ...	ABC 123 0	Then	ABC 123 On Time
Else If	Arrival Delay	is less than or equ...	ABC 123 15	Then	ABC 123 Minor Delay
Else If	Arrival Delay	is less than or equ...	ABC 123 60	Then	ABC 123 Moderate Delay ...

Add Clause

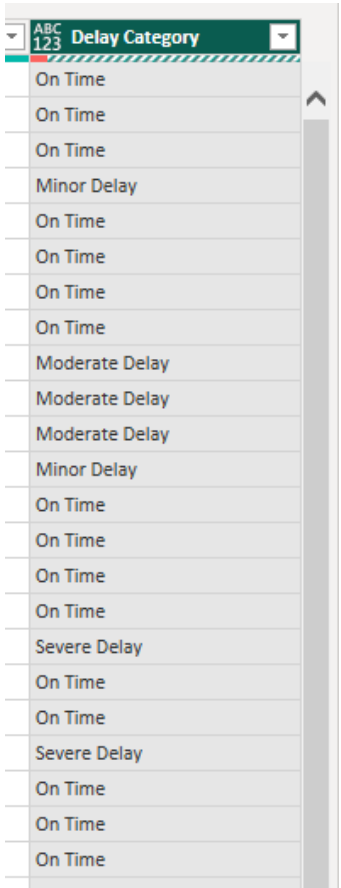
Else ①
ABC 123 Severe Delay

OK Cancel

"Delay Category" built from Arrival Delay: On Time / Minor Delay / Moderate Delay / Severe Delay.

3.3 Create a Conditional Column (cont.)



A screenshot of a data table interface. The table has a header row with a green background and a dropdown arrow on the left. The header text is 'ABC 123 Delay Category'. Below the header, there are 25 rows of data. The 'Delay Category' column contains the following values from top to bottom: On Time, On Time, On Time, Minor Delay, On Time, On Time, On Time, On Time, On Time, Moderate Delay, Moderate Delay, Moderate Delay, Minor Delay, On Time, On Time, On Time, On Time, Severe Delay, On Time, On Time, Severe Delay, On Time, On Time, On Time.

ABC 123 Delay Category
On Time
On Time
On Time
Minor Delay
On Time
On Time
On Time
On Time
On Time
Moderate Delay
Moderate Delay
Moderate Delay
Minor Delay
On Time
On Time
On Time
On Time
Severe Delay
On Time
On Time
Severe Delay
On Time
On Time
On Time

Delay Category column populated as expected.

3.4 Extract Date Parts

1 ² ₃ Year	1 ² ₃ Month	1 ² ₃ Day	1 ² ₃ Quarter
2011	1	1	1
2011	1	2	1
2011	1	3	1
2011	1	4	1
2011	1	5	1
2011	1	6	1
2011	1	7	1
2011	1	8	1
2011	1	9	1
2011	1	10	1
2011	1	11	1

Year, Month, Day, and Quarter extracted from Flight Date into separate columns.

4. Advanced Tasks

4.1 Merge Queries

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Merge

Select tables and matching columns to create a merged table.

hflights

Flight Date	Week Day	Departure Time	Arrival Time	Airline	Flight Number	Aircraft ID	ActualElaps
01/01/2011	6	1400	1500	AA	428	N576AA	
02/01/2011	7	1401	1501	AA	428	N557AA	
03/01/2011	1	1352	1502	AA	428	N541AA	
04/01/2011	2	1403	1513	AA	428	N403AA	
05/01/2011	3	1405	1503	AA	428	N403AA	

hflights (2)

Flight Date	Week Day	Departure Time	Arrival Time	Airline	Flight Number	Aircraft ID	ActualElaps
01/01/2011	6	1400	1500	AA	428	N576AA	
02/01/2011	7	1401	1501	AA	428	N557AA	
03/01/2011	1	1352	1502	AA	428	N541AA	
04/01/2011	2	1403	1513	AA	428	N403AA	
05/01/2011	3	1405	1503	AA	428	N403AA	

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

✓ The selection matches 227496 of 227496 rows from the first table.

OKCancel

hflights merged with hflights (2), Left Outer join — 227,496 of 227,496 rows matched.

4.2 Append Queries

Append

Concatenate rows from two tables into a single table.

☒ Two tables ☐ Three or more tables

First table

hflights

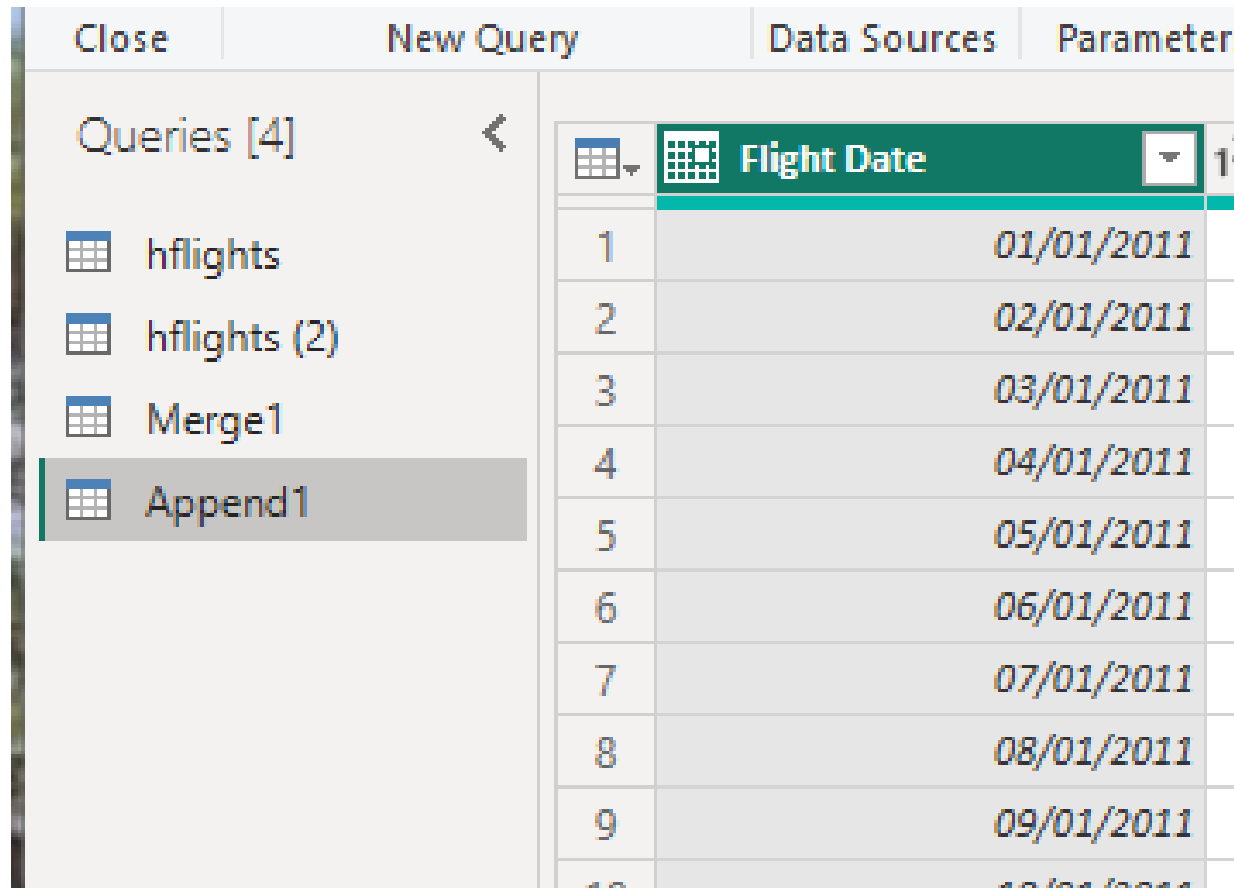
Second table

hflights (2)

OK Cancel

hflights and hflights (2) appended (stacked) into one table.

4.2 Append Queries (cont.)

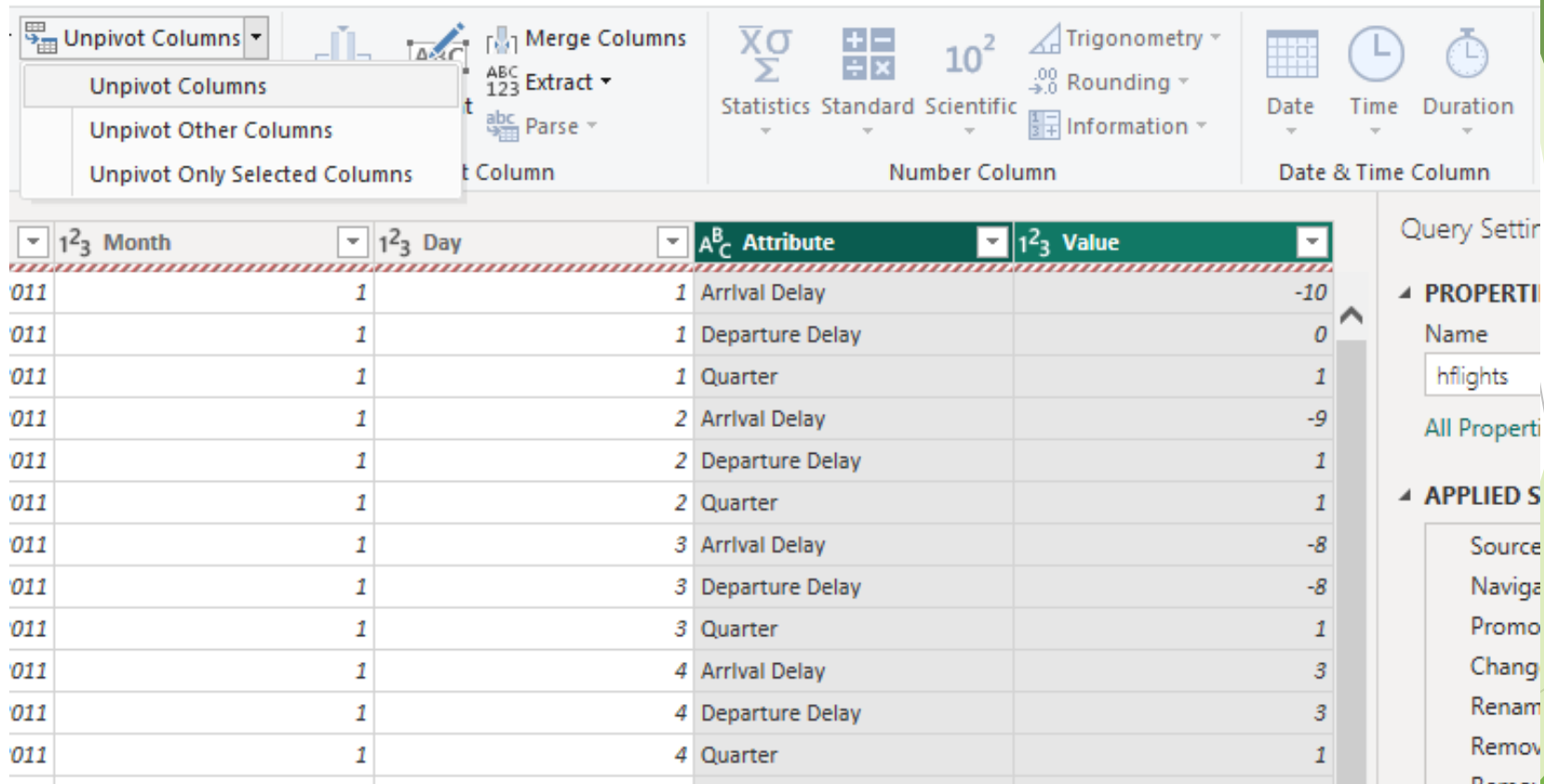


The screenshot shows a software interface with a top navigation bar containing 'Close', 'New Query', 'Data Sources', and 'Parameters'. On the left, a sidebar titled 'Queries [4]' with a back arrow contains a list of queries: 'hflights', 'hflights (2)', 'Merge1', and 'Append1'. The 'Append1' query is selected and highlighted. The main area displays a table with a header row 'Flight Date' and a body of 10 rows, each containing a number from 1 to 10 and a date from '01/01/2011' to '10/01/2011'.

	Flight Date
1	01/01/2011
2	02/01/2011
3	03/01/2011
4	04/01/2011
5	05/01/2011
6	06/01/2011
7	07/01/2011
8	08/01/2011
9	09/01/2011
10	10/01/2011

New Append1 query confirms the combined table.

4.3 Unpivot Columns

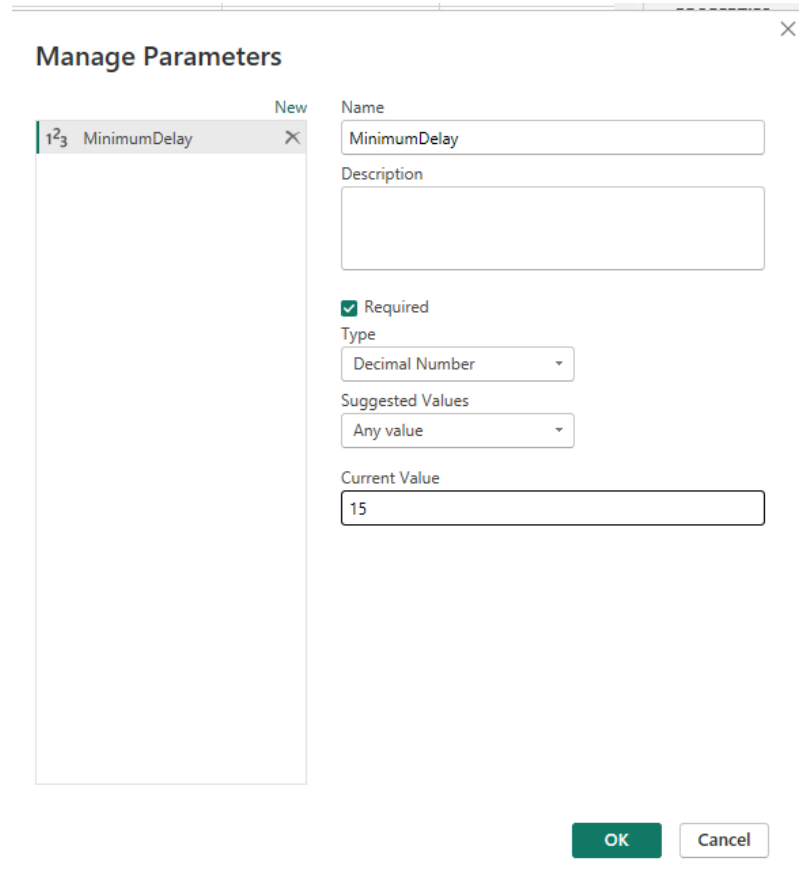


The screenshot shows the Power BI Desktop interface. The 'Unpivot Columns' menu is open, displaying three options: 'Unpivot Columns', 'Unpivot Other Columns', and 'Unpivot Only Selected Columns'. The background table has four columns: 'Month', 'Day', 'Attribute', and 'Value'. The 'Attribute' column contains values like 'Arrival Delay', 'Departure Delay', and 'Quarter', while the 'Value' column contains numerical values. The 'Query Settings' pane on the right shows the 'PROPERTY' tab with 'Name' set to 'hflights'.

Month	Day	Attribute	Value
011	1	Arrival Delay	-10
011	1	Departure Delay	0
011	1	Quarter	1
011	1	Arrival Delay	-9
011	1	Departure Delay	1
011	1	Quarter	1
011	1	Arrival Delay	-8
011	1	Departure Delay	-8
011	1	Quarter	1
011	1	Arrival Delay	3
011	1	Departure Delay	3
011	1	Quarter	1

Unpivot Columns used to reshape wide columns into Attribute/Value pairs.

4.4 Create New Parameters



The screenshot shows a 'Manage Parameters' dialog box. On the left, a list contains one item: '1²₃ MinimumDelay'. To the right of this list is a 'New' button. The main area of the dialog contains the following fields and controls:

- Name:** A text box containing 'MinimumDelay'.
- Description:** An empty text box.
- Required:** A checked checkbox.
- Type:** A dropdown menu showing 'Decimal Number'.
- Suggested Values:** A dropdown menu showing 'Any value'.
- Current Value:** A text box containing '15'.

At the bottom right of the dialog are 'OK' and 'Cancel' buttons.

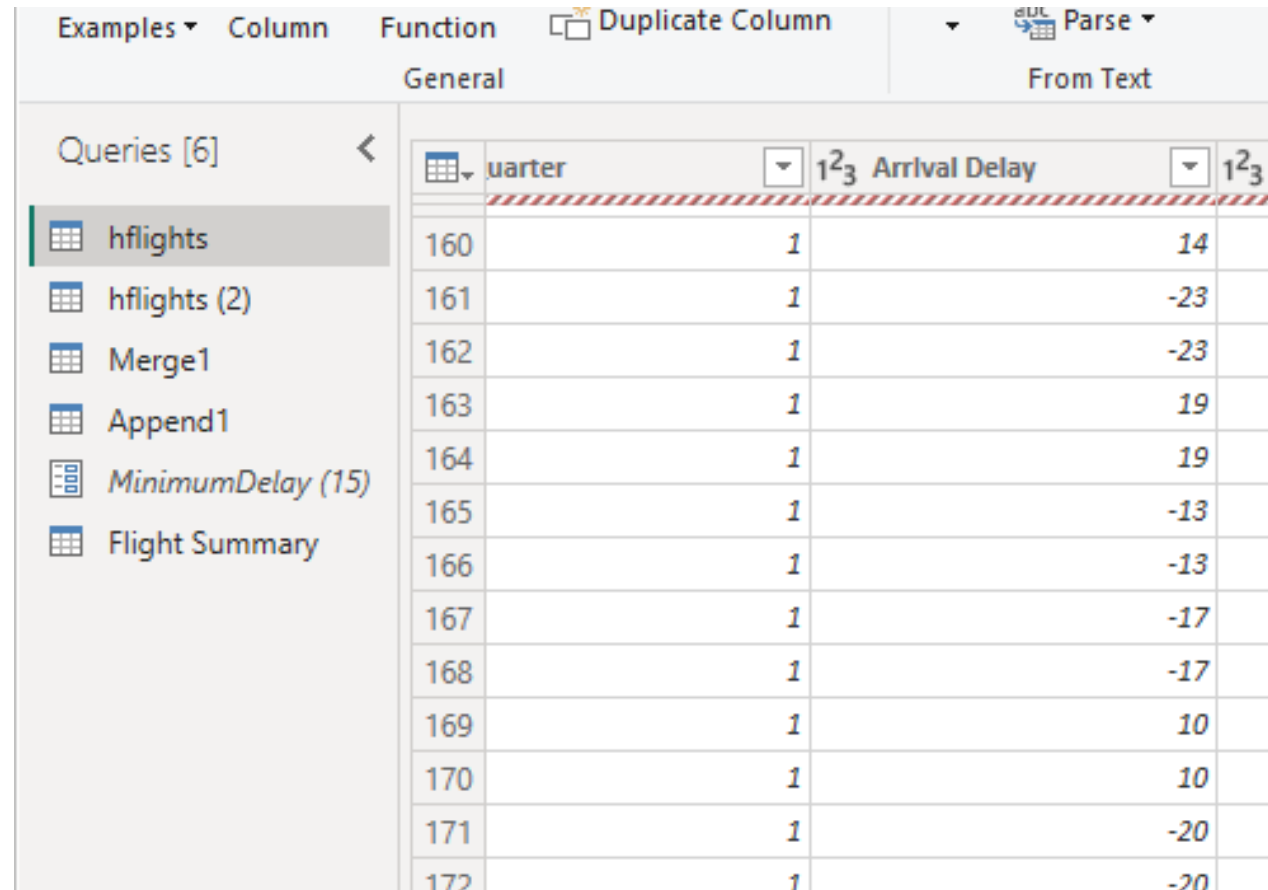
Parameter "MinimumDelay" created — Decimal Number, current value 15.

4.4 Create New Parameters (cont.)

The screenshot displays a software interface with a tabbed menu at the top: 'Close', 'New Query', 'Data Sources', 'Parameters', 'Output Data', and 'Query'. The 'Parameters' tab is active. On the left, a 'Queries [5]' sidebar lists five items: 'hflights', 'hflights (2)', 'Merge1', 'Append1', and 'MinimumDelay (15)'. The 'MinimumDelay (15)' item is highlighted with a grey background. The main workspace on the right shows the 'Current Value' of the selected parameter as '15' in a text input field. Below this field is a button labeled 'Manage Parameter'.

MinimumDelay parameter appears in the Queries list, ready to be referenced.

4.5 Create a Reference Query



The screenshot shows a data tool interface. On the left, a sidebar titled 'Queries [6]' lists several queries: 'hflights' (selected), 'hflights (2)', 'Merge1', 'Append1', 'MinimumDelay (15)', and 'Flight Summary'. On the right, a data table is displayed with the following columns: 'Quarter', 'Arrival Delay', and 'Flight ID'. The table contains 13 rows of data, with flight IDs ranging from 160 to 172. The 'Arrival Delay' column shows various values, including positive and negative numbers.

Quarter	Arrival Delay	Flight ID
1	14	160
1	-23	161
1	-23	162
1	19	163
1	19	164
1	-13	165
1	-13	166
1	-17	167
1	-17	168
1	10	169
1	10	170
1	-20	171
1	-20	172

"Flight Summary" created as a reference query, built on an existing query's output.

4.6 Nested Conditional Column

×

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name

Distance Category

	Column Name	Operator	Value ⓘ		Output ⓘ	
If	Distance ▾	is less than ▾	ABC 123 ▾ 300	Then	ABC 123 ▾ short Haul	...
Else If	Distance ▾	is less than ▾	ABC 123 ▾ 1000	Then	ABC 123 ▾ Medium Haul	...

Add Clause

Else ⓘ

ABC 123 ▾ long Haul



OK

Cancel

"Distance Category" built from Distance: short Haul / Medium Haul / long Haul.

4.6 Column distribution, column profiling, column quality

Flight Date

Valid

100%

Error

0%

Empty

0%

331 distinct, 67 unique

Week Day

Valid

100%

Error

0%

Empty

0%

7 distinct, 0 unique

Departure Time

Valid

100%

Error

0%

Empty

0%

205 distinct, 62 unique

Arrival Time

Valid

100%

Error

0%

Empty

0%

504 distinct, 253 unique

Airline

Valid

100%

Error

0%

Empty

0%

1 distinct, 0 unique

Flight Number

Valid

100%

Error

0%

Empty

0%

410 distinct, 206 unique

Aircraft ID

Valid

100%

Error

0%

Empty

0%

246 distinct, 38 unique

ActualElapsedTime

Valid

100%

Error

0%

Empty

0%

96 distinct, 4 unique

AirTime

Valid

100%

Error

0%

Empty

0%

95 distinct, 7 unique

	Flight Date	Week Day	Departure Time	Arrival Time	Airline	Flight Number	Aircraft ID	ActualElapsedTime	AirTime
1	24/10/2011			1340	ExpressJet		4092		131
2	12/10/2011			1317	ExpressJet		4103		110
3	29/10/2011			1119	ExpressJet		4172		79
4	21/10/2011			1127	ExpressJet		4202		96
5	25/10/2011			1427	ExpressJet		4259		115
6	25/10/2011			1527	ExpressJet		4283		132
7	25/10/2011			1302	ExpressJet		4299		152
8	30/10/2011			1412	ExpressJet		4321		94
9	22/10/2011			854	ExpressJet		4337		123
10	17/10/2011			1802	ExpressJet		4361		98
11	22/10/2011			717	ExpressJet		4379		165
12	11/10/2011			717	ExpressJet		4379		167
13	05/10/2011			1357	ExpressJet		4431		136
14	13/10/2011			1447	ExpressJet		4432		114
15	27/10/2011			1412	ExpressJet		4457		117
16	21/10/2011			1412	ExpressJet		4457		117
17	14/10/2011			1140	ExpressJet		4486		112
18	10/10/2011			1327	ExpressJet		4607		119
19	25/11/2011			717	ExpressJet		4159		130

Column statistics

Count

1000

Error

0

Empty

0

Distinct

331

Unique

67

Min

02/01/2...

Max

30/12/2...

Average

28/06/2...

Value distribution

12/12/2011

02/08/2011

18/03/2011

07/07/2011

14/01/2011

03/01/2011

13/09/2011

10/02/2011

14/10/2011

16/02/2011

06/06/2011

28/03/2011

08/01/2011

08/05/2011

01/06/2011

06/05/2011

22/09/2011

25/02/2011

02/01/2011

24/10/2011

04/07/2011

29/01/2011

26/01/2011

25/09/2011

19/12/2011

25/03/2011

06/04/2011

21/10/2011

29/10/2011

07/10/2011

01/03/2011

05/08/2011

20/03/2011

18/11/2011

23/01/2011

02/04/2011

27/07/2011

28/04/2011

20/02/2011

24/08/2011

17/02/2011

04/08/2011

13/07/2011

12/09/2011

23/11/2011

24/07/2011

06/03/2011

13/11/2011

06/07/2011

04/11/2011

Query Settings

PROPERTIES

Name

cleaned_dataset (3)

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

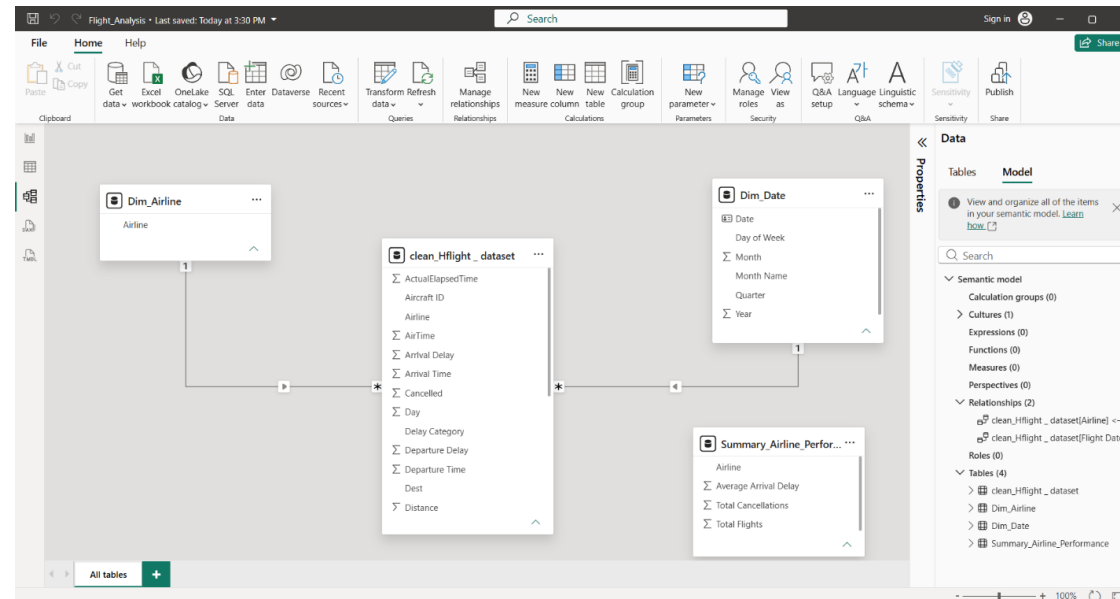
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Part 3: Data Modelling

Data Modelling

Star Schema design pattern



The Power BI data model for the Airline Delay and Operations Dashboard is constructed using a standard Star Schema design pattern. The model separates quantitative transactional data (facts) from descriptive background details (dimensions) to ensure optimal query performance, clear data relationships, and intuitive reporting.

The Model — Four Primary Tables

- clean_Hflight_dataset (Central Fact Table)
- Dim Airline (Dimension Table)
- Dim_Date (Dimension Table)
- Summary_Airline_Performance (Aggregated Table)

Table Classification & Structural Interpretation

A. Central Fact Table: clean_Hflight_dataset

- Classification: Fact Table.
- Interpretation: This table serves as the primary engine for analysis. It stores individual flight record details and transactional metrics. It contains numerical fields required to calculate Key Performance Indicators (KPIs), such as Arrival Delay, Departure Delay, AirTime, Distance, ActualElapsedTime, and operational flags such as Cancelled.

Dimension Tables: Dim_Airline & Dim_Date

- Classification: Dimension Tables (Lookup Tables).
- Interpretation: These tables provide descriptive attributes that allow users to filter, group, and slice the flight metrics stored in the Fact table.
- Dim_Airline: Contains distinct airline identifiers (Airline), allowing the dashboard to break down performance, delays, and cancellations by carrier.
- Dim_Date: Contains detailed temporal attributes (Date, Day of Week, Month, Month Name, Quarter, Year). This enables trend analysis, identifying peak travel periods, seasonal delay patterns, and month-over-month performance.

Standalone Summary Table: Summary_Airline_Performance

- Classification: Aggregated Reference Table.
- Interpretation: Contains pre-calculated summary metrics (Average Arrival Delay, Total Cancellations, Total Flights) per airline. It acts as an auxiliary table for high-level summary cards or quick verification against detailed DAX measures.

Data Relationships and Cardinality

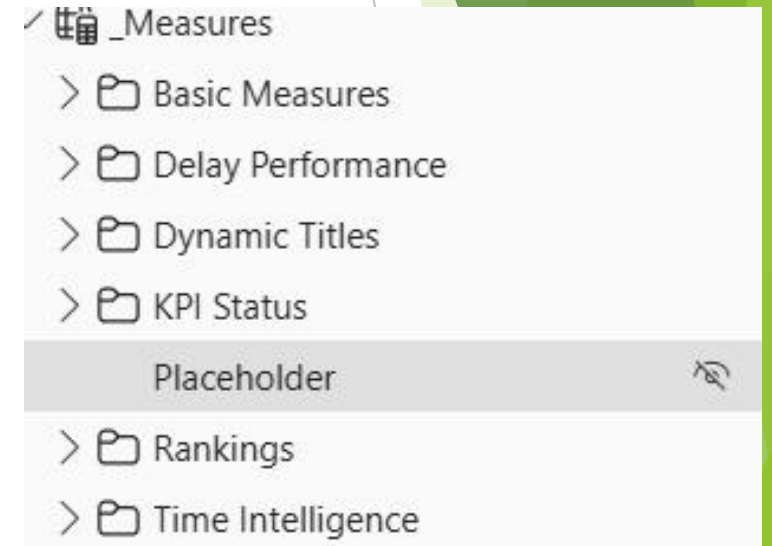
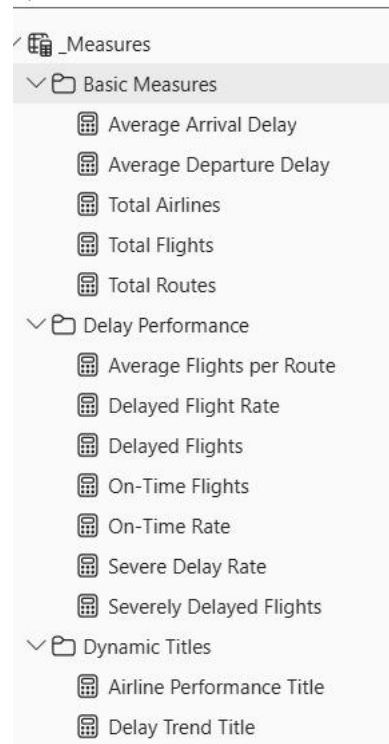
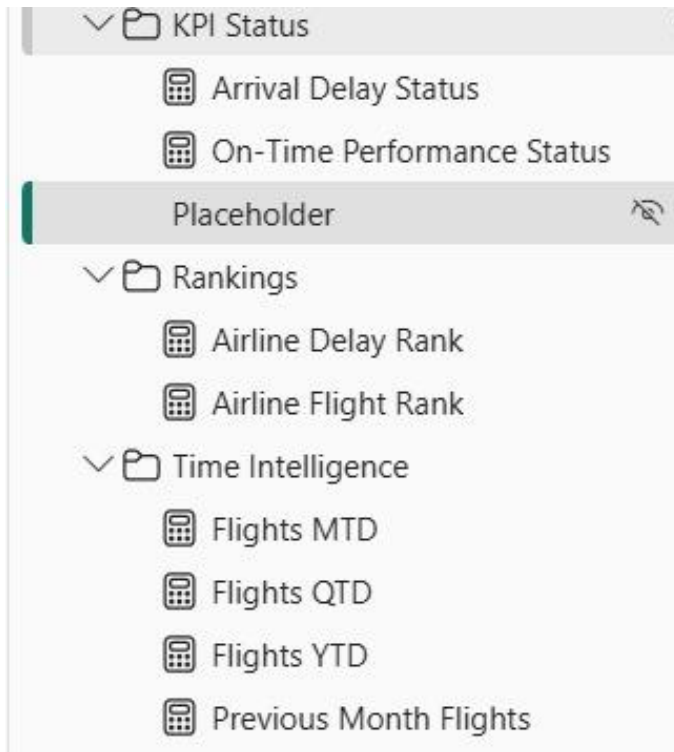
The integrity of the model is maintained through proper database relationships

- Dim_Airline to clean_Hflight_dataset — Cardinality: One-to-Many (1:*). Direction: single-direction filter flowing from Dim_Airline to clean_Hflight_dataset. Meaning: a single airline in the dimension table links to multiple flight records in the fact table; filtering by an airline automatically isolates all relevant flights for that carrier.
- Dim_Date to clean_Hflight_dataset — Cardinality: One-to-Many (1:*). Direction: single-direction filter flowing from Dim_Date to clean_Hflight_dataset. Meaning: a unique calendar date connects to multiple flights occurring on that day; selecting a month, quarter, or date range dynamically filters all operational metrics in the fact table.

Part 4: DAX Measures

DAX Measures

We created and tested 22 DAX measures covering flight volumes, delay performance, time intelligence, percentages, rankings, KPI statuses and dynamic titles. The measures were organised into display folders and prepared for use in KPI cards, trend charts, ranking tables, slicers and drill-through analysis.

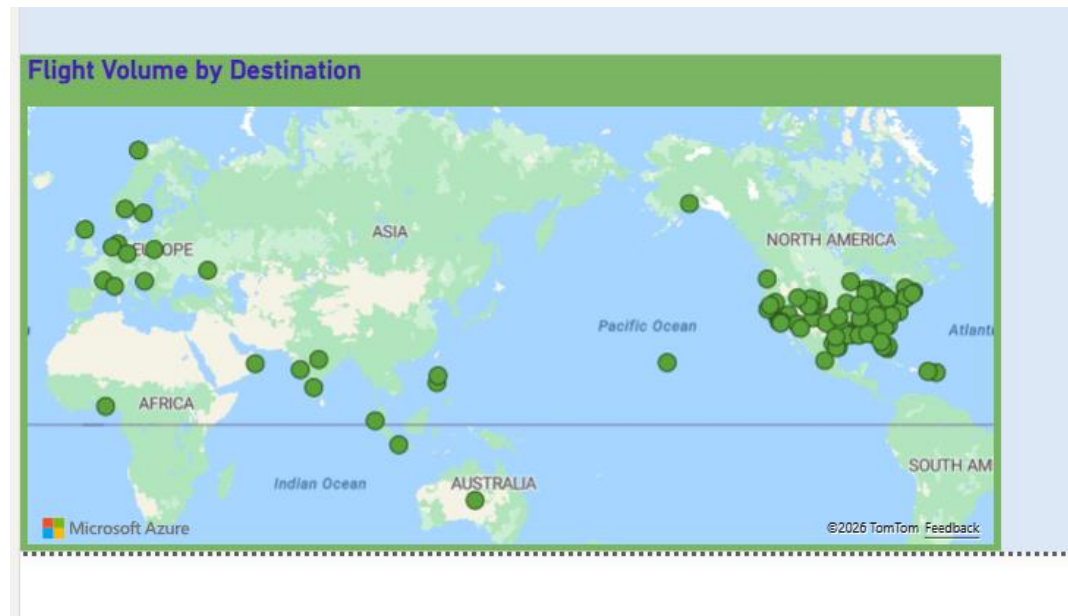


Part 5: Advanced Dashboard Design and Visuals

Airline Delay and Operations Dashboard

1. Geographic and Segment Analysis — Flight Volume by Destination

My section focused on analysing flight destinations, airline performance, monthly activity and delay patterns using Power BI. The map was used to show how flights are distributed across different destinations. The results show that flight activity is concentrated in certain areas, particularly North America. Some destinations have much higher flight volumes than others.



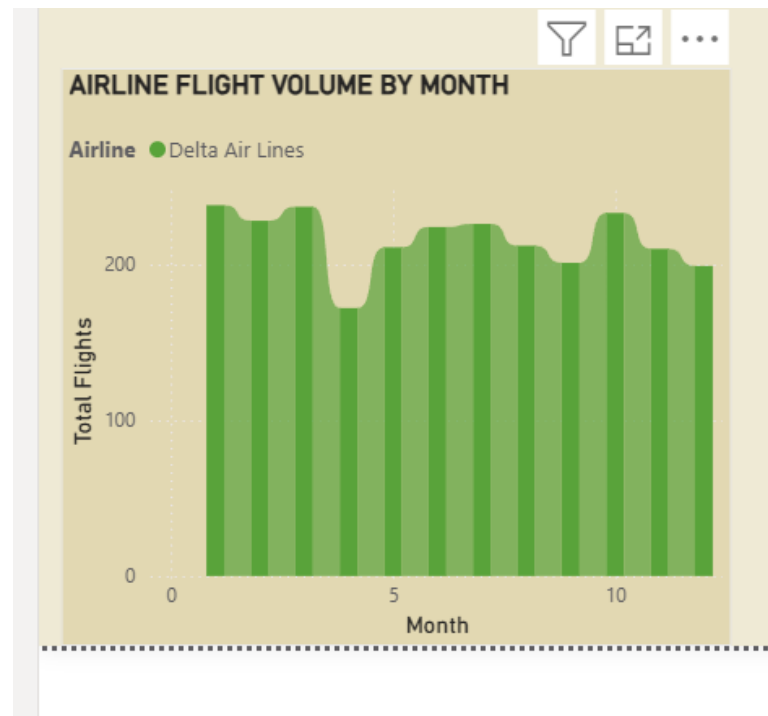
Airline Performance by Month

The matrix compares airlines across months using Total Flights, Delayed Flights and Severe Delay Rate. For example, in Month 1, Continental Airlines recorded approximately 6,001 flights, while Southwest Airlines recorded 3,667 flights. Southwest had a severe delay rate of 3.9%, compared with 2.5% for Continental.

Airline Performance by Month				
Month	1		2	
Airline	Total Flights	Severe Delay Rate	Delayed Flights	Total F
AirTran Airways	133	2.3%	50	
Alaska Airlines	30	3.3%	18	
American Airlines	270	2.6%	90	
Continental Airlines	6,001	2.5%	2,831	
Delta Air Lines	238	3.8%	92	
Envoy Air	325	5.8%	131	
ExpressJet	6,807	2.3%	3,010	
Frontier Airlines	62	1.6%	47	
JetBlue Airways	48	10.4%	17	
Mesa Airlines				
SkyWest Airlines	689	3.3%	355	
Southwest Airlines	3,667	3.9%	1,974	
Total	18,665	2.8%	8,728	1

Monthly Flight Activity

The ribbon chart was used to examine changes in airline flight activity across months. It shows that flight volumes vary throughout the year, helping identify possible seasonal travel patterns.



2. Detailed Airline Analysis

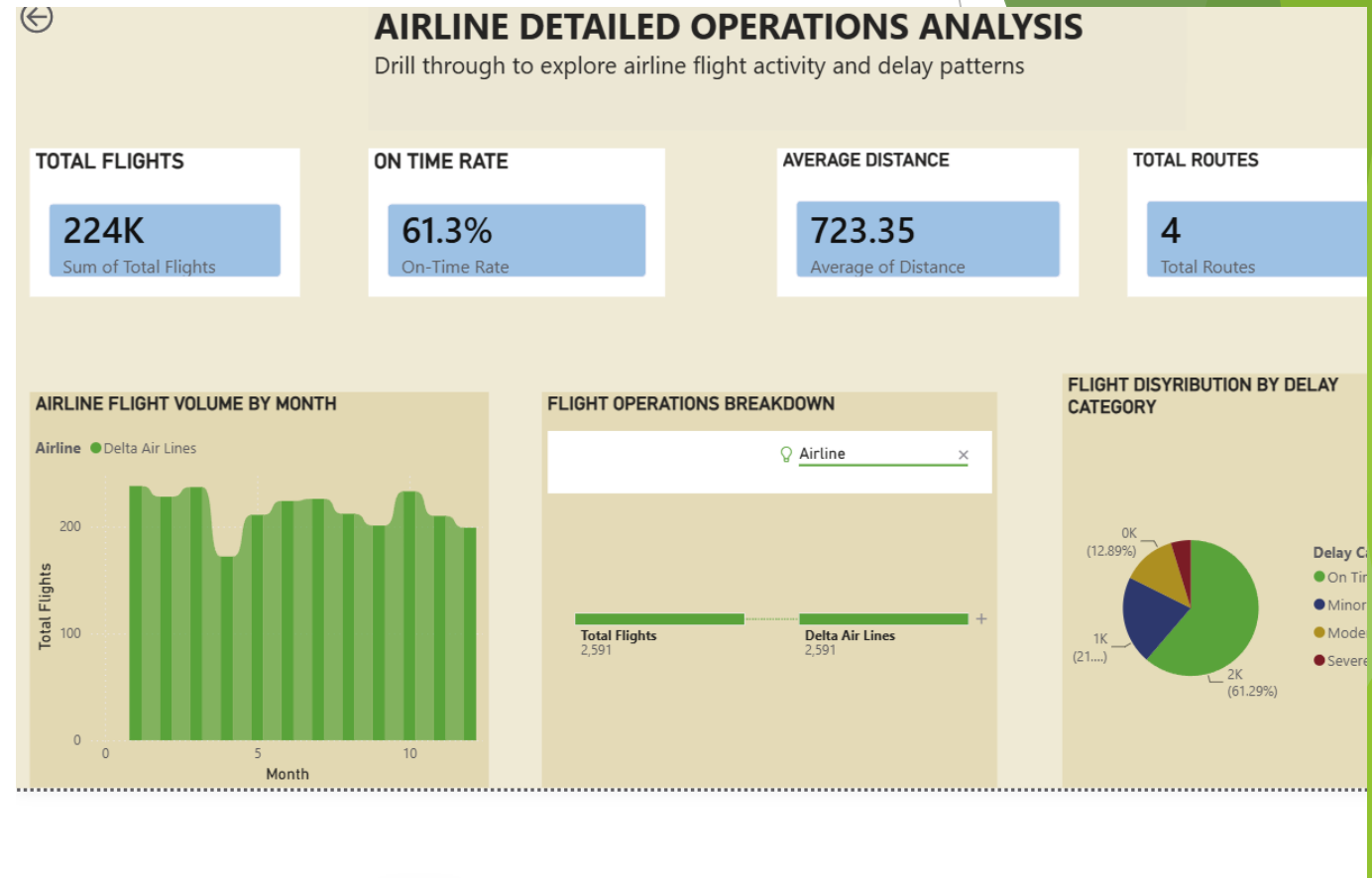
Detailed Airline Analysis — Drill-through

- A drill-through page was created to allow users to select an airline and view its detailed performance.

Delta Air Lines — Dashboard Summary

- Total Flights: 224K
- On-Time Rate: 61.3%
- Average Distance: 723.35
- Total Routes: 4

The on-time rate of 61.3% indicates that there is still room for improvement in airline schedule reliability.

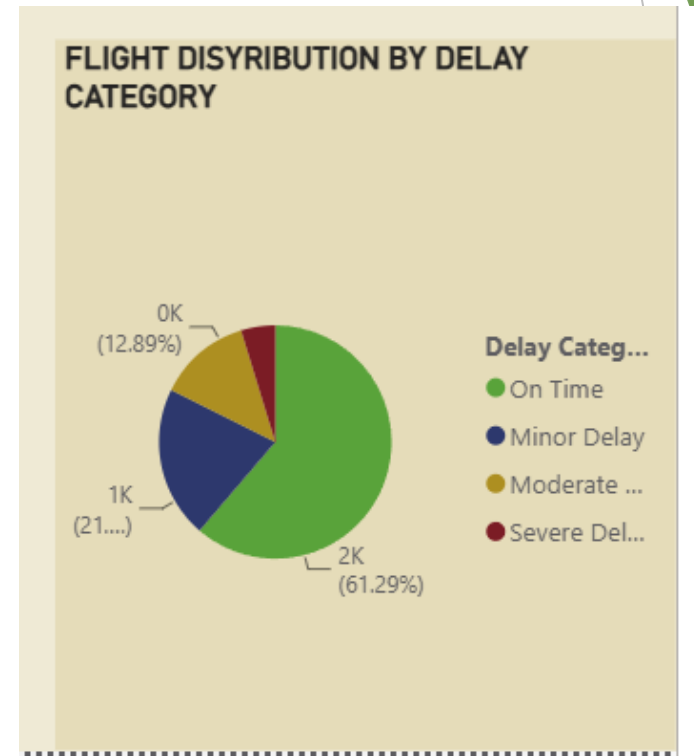


3. Delay Category Analysis

The pie chart shows Delta's flight distribution by delay category

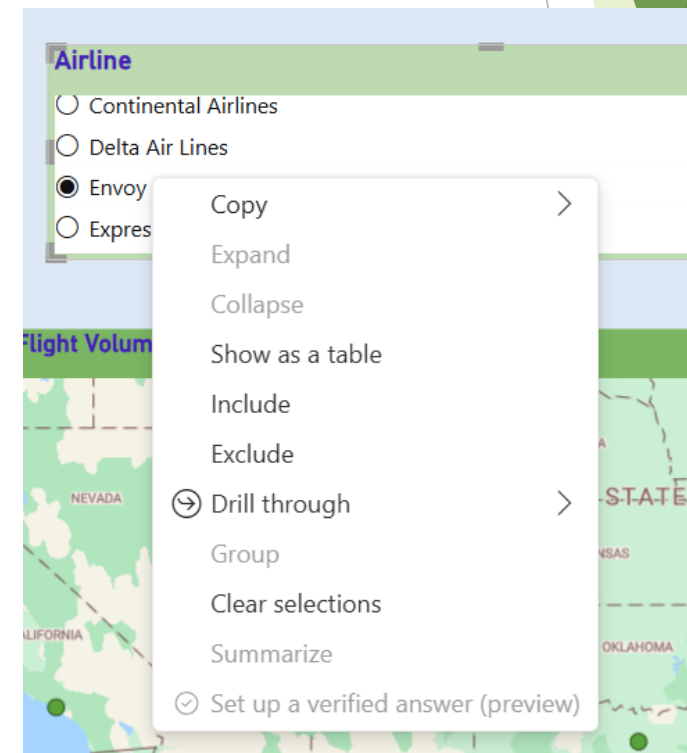
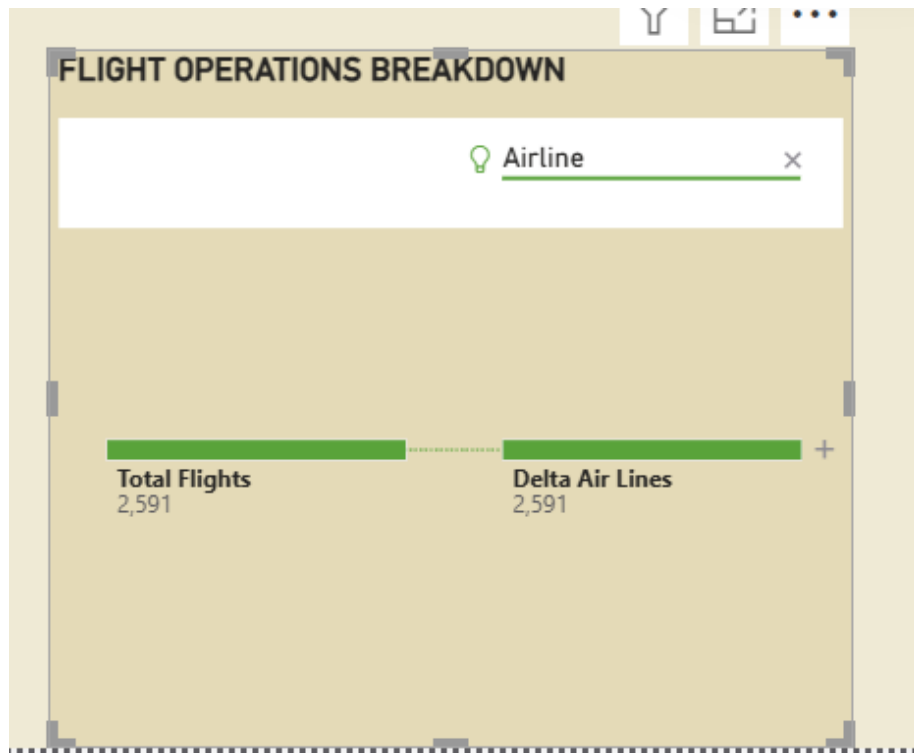
- On Time: 61.29%
- Minor Delay: approximately 21%
- Moderate Delay: approximately 12.9%
- Severe Delay: approximately 4%

The majority of flights were on time, while severe delays represented the smallest category.



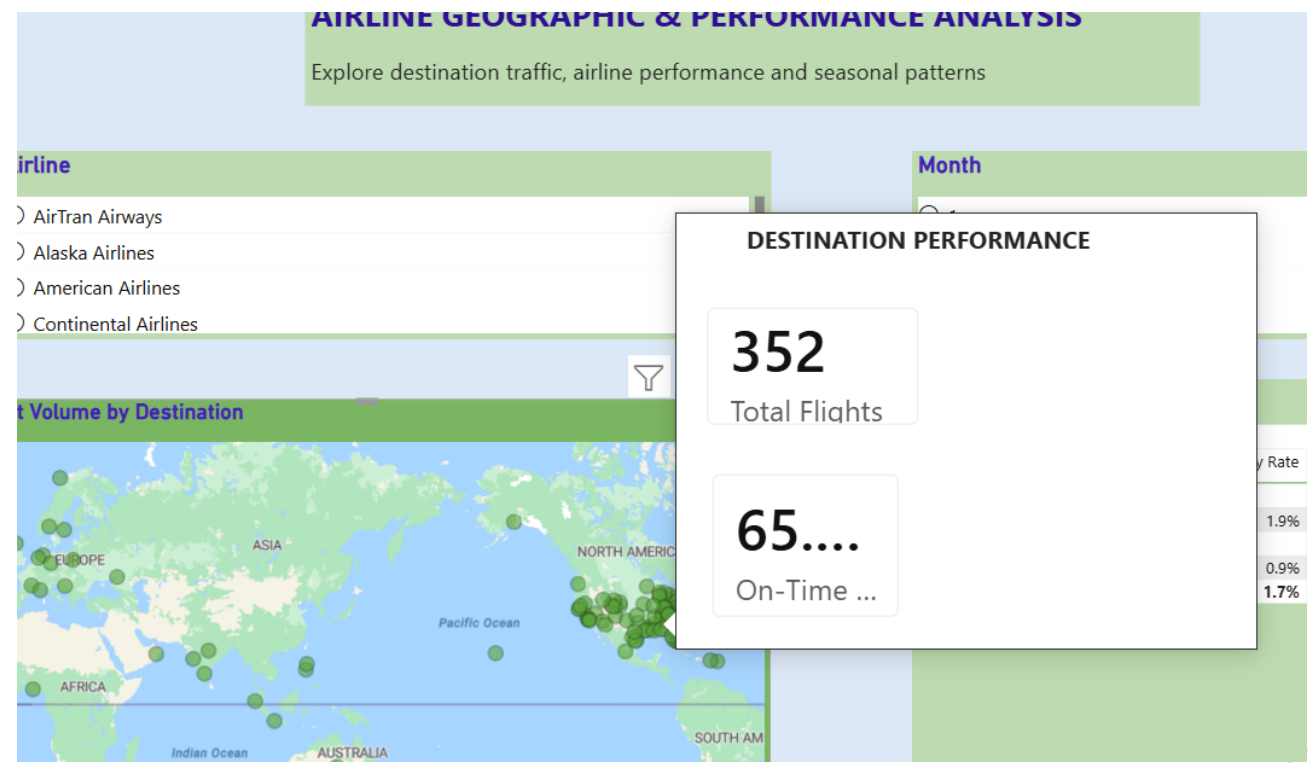
4. Decomposition Tree and Drill-through

The decomposition tree allows total flights to be broken down by airline, destination, month, flight status and delay category. This makes it easier to investigate the factors behind flight activity. The drill-through feature allows users to select an airline from the main dashboard and automatically move to a detailed airline analysis page.



5. Custom Tooltip

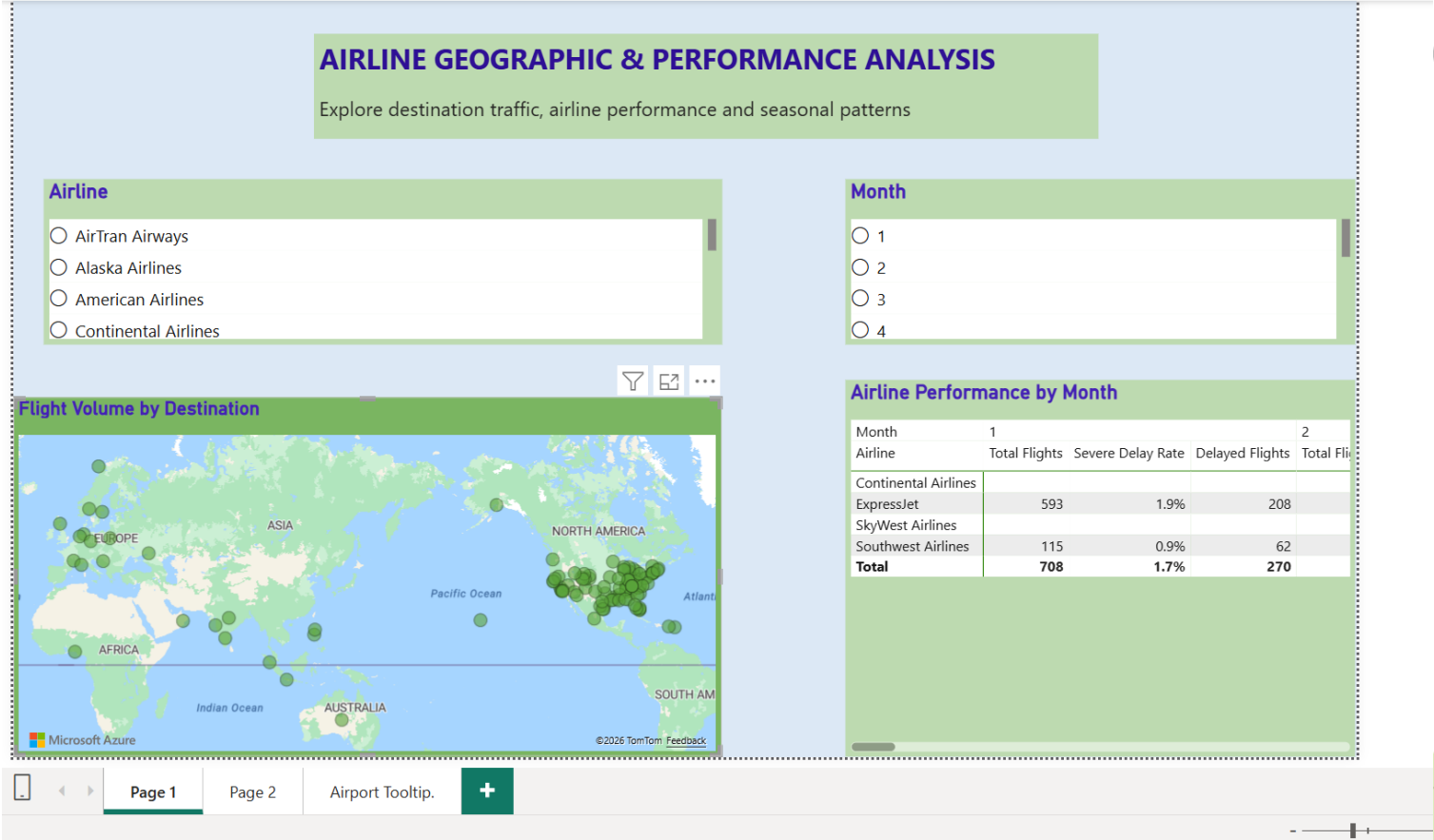
A custom tooltip was added to provide additional information when users hover over a visual without overcrowding the dashboard.



Conclusion

Overall, the dashboard shows that airline operations vary by destination, airline and month. Flight volume is concentrated in certain destinations, while airline performance and delay rates differ between carriers. The interactive drill-through, decomposition tree and tooltip allow users to move from general information to detailed airline analysis and identify areas that may require improvement.

Dashboard Visuals — Figure 1: Geographic and Segment Analysis



This page presents the geographic distribution of flight activity and allows airline performance to be compared across different months.

Figure 2: Detailed Airline Operations — Page 4

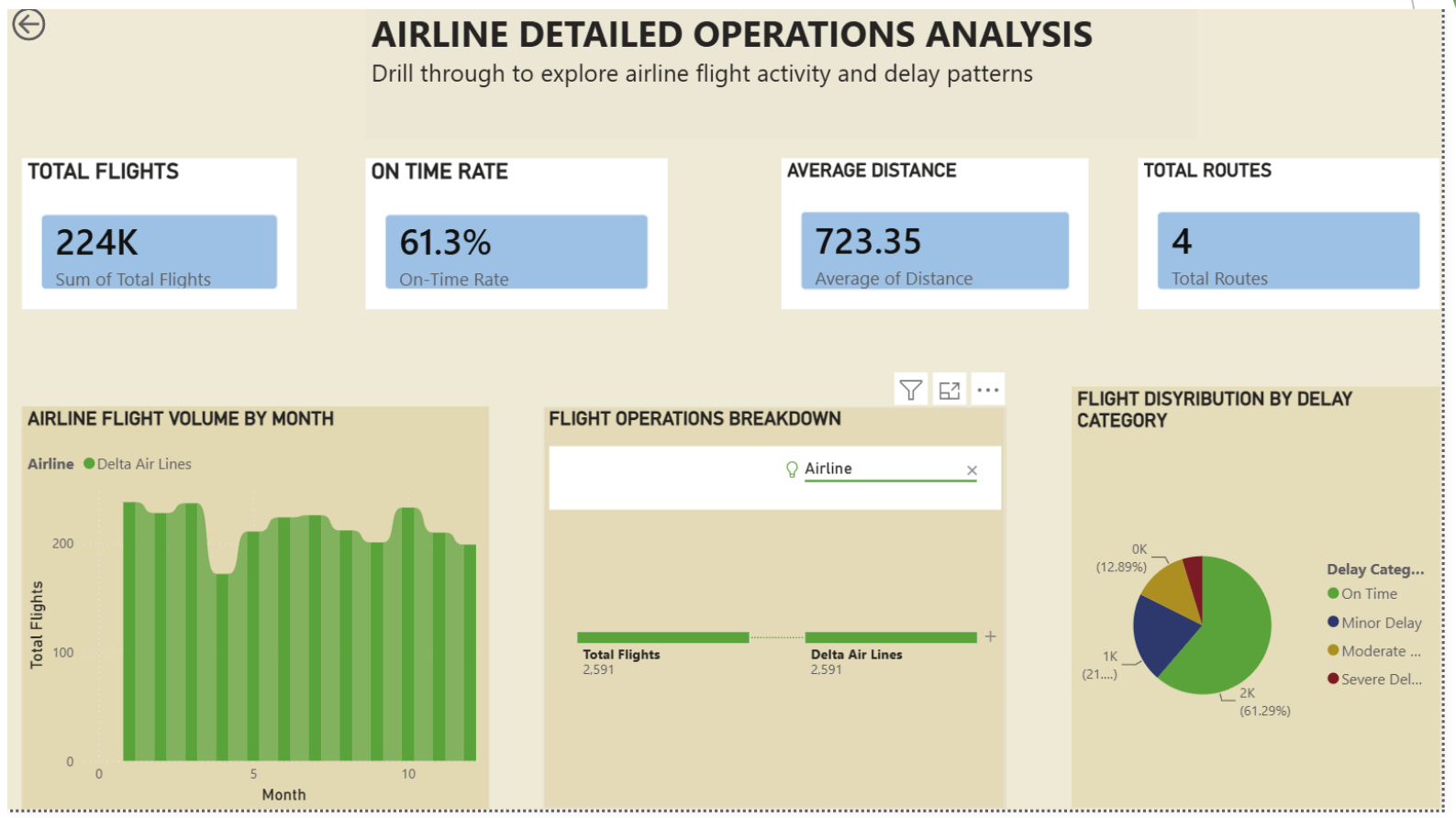


Figure 2: Airline Detailed Operations Analysis Dashboard. This page provides detailed information about the selected airline, including total flights, on-time rate, average distance, routes, monthly activity and delay categories.

Figure 3: Custom Tooltip

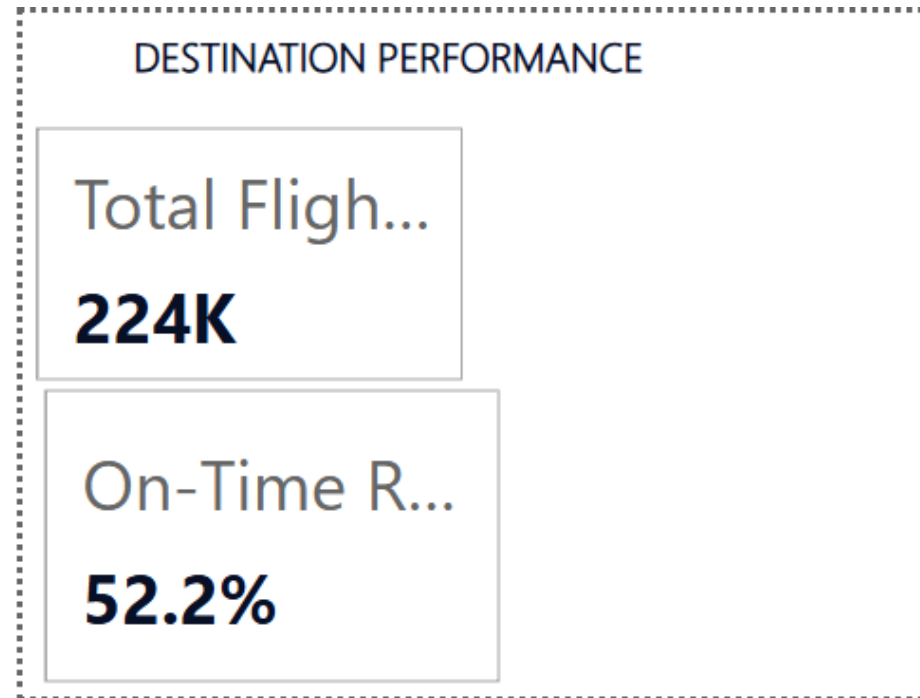
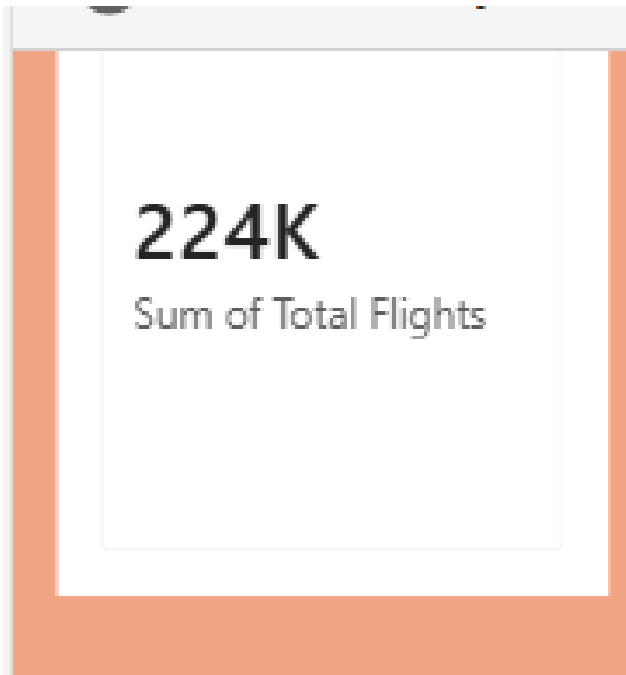


Figure 3: Custom Tooltip Showing Additional Destination Performance Information

AirTran Airways — Q1 2011 Total Flights

For AirTran Airways in the first quarter of 2011, the dashboard recorded a total of 224,000 flights. This is the baseline volume figure used for the other Executive Summary measures in this filtered view.



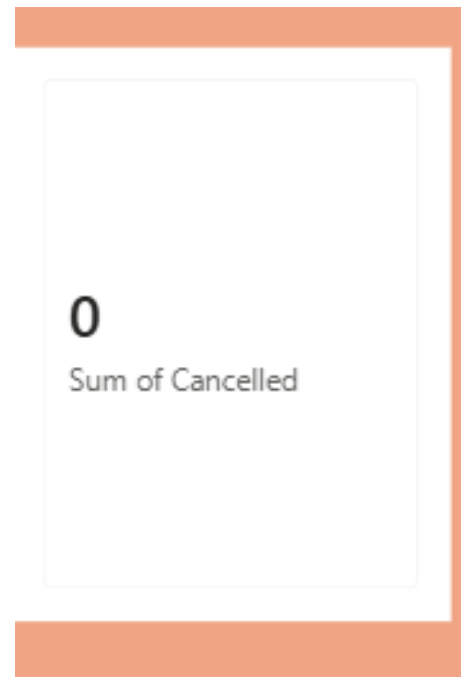
Insight 2: Average Departure and Arrival Delay

The average departure delay for AirTran Airways in Q1 2011 is 6.86 minutes, while the average arrival delay is lower at 5.73 minutes. This gap suggests that although flights tend to leave a few minutes late on average, the airline generally makes up a small amount of that time in the air, arriving slightly closer to schedule than it departs.



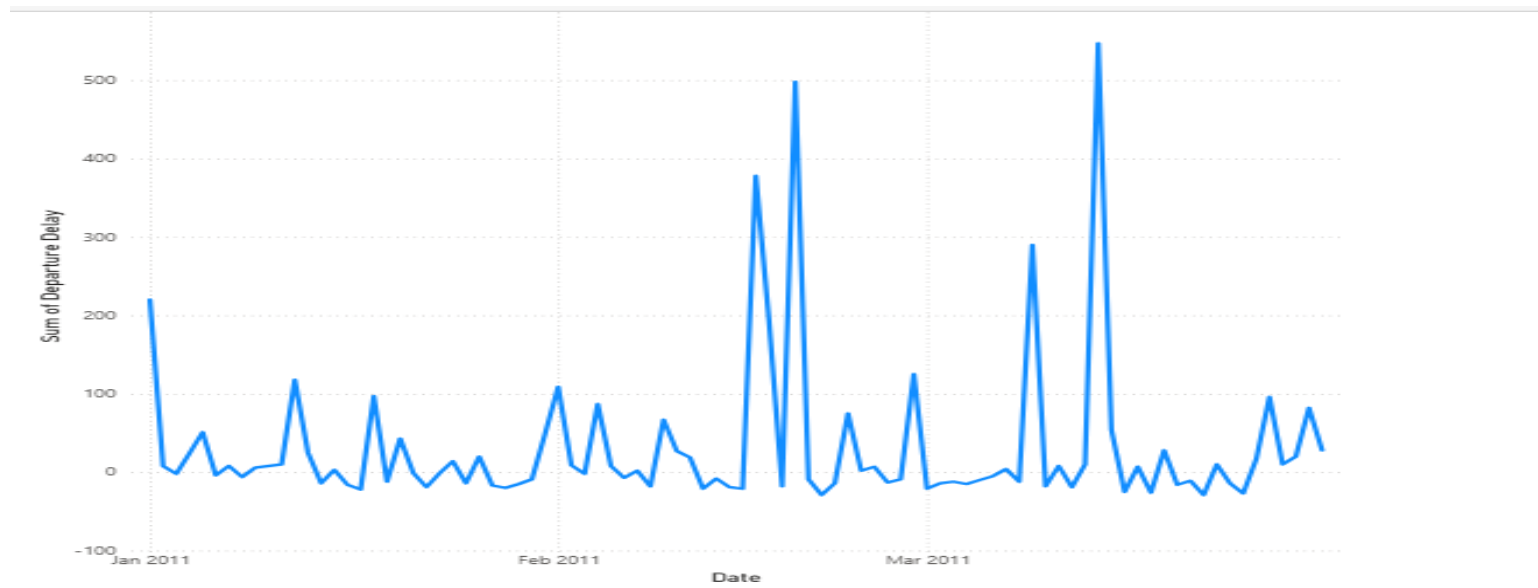
Insight 3: Cancellations

The dashboard shows 0 cancelled flights for AirTran Airways in this period. This is a strong result, indicating that during Q1 2011 the airline maintained full operational reliability with no recorded cancellations, at least within the filtered view.



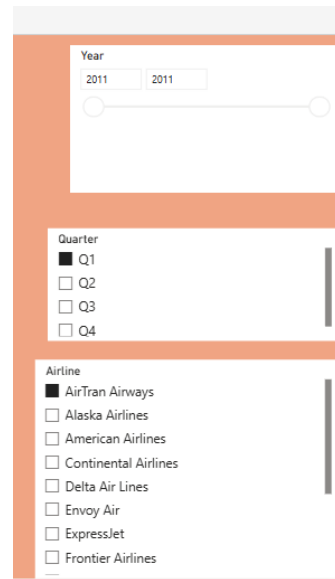
Insight 4: Delay Spikes Over Time

The trend chart for January to March 2011 shows that daily departure delays generally stay low, fluctuating between 0 and 100 minutes on most days, aside from an early spike near 220 minutes in January. From February onward, several sharp spikes stand out, reaching roughly 380, 500, and 545 minutes on individual days, with the highest spike occurring in March. This pattern points to isolated severe-delay days rather than a steady elevated delay level, meaning the 6.86-minute average delay figure alone hides these extreme outlier days.



Insight 5: Scope of Current View

All figures above reflect AirTran Airways only, filtered to Q1 2011, since these were the only Airline and Quarter selections active on the dashboard when these values were captured. A full, unfiltered view across all airlines and all quarters is needed to confirm whether AirTran Airways' performance is typical or an outlier compared to the rest of the dataset.



The image shows a vertical filter panel from a dashboard. It contains three sections: 'Year' with a range slider set to 2011, 'Quarter' with radio buttons for Q1, Q2, Q3, and Q4 (Q1 is selected), and 'Airline' with checkboxes for AirTran Airways, Alaska Airlines, American Airlines, Continental Airlines, Delta Air Lines, Envoy Air, ExpressJet, and Frontier Airlines (AirTran Airways is selected).

Year

2011 2011

Quarter

☒ Q1

☐ Q2

☐ Q3

☐ Q4

Airline

☒ AirTran Airways

☐ Alaska Airlines

☐ American Airlines

☐ Continental Airlines

☐ Delta Air Lines

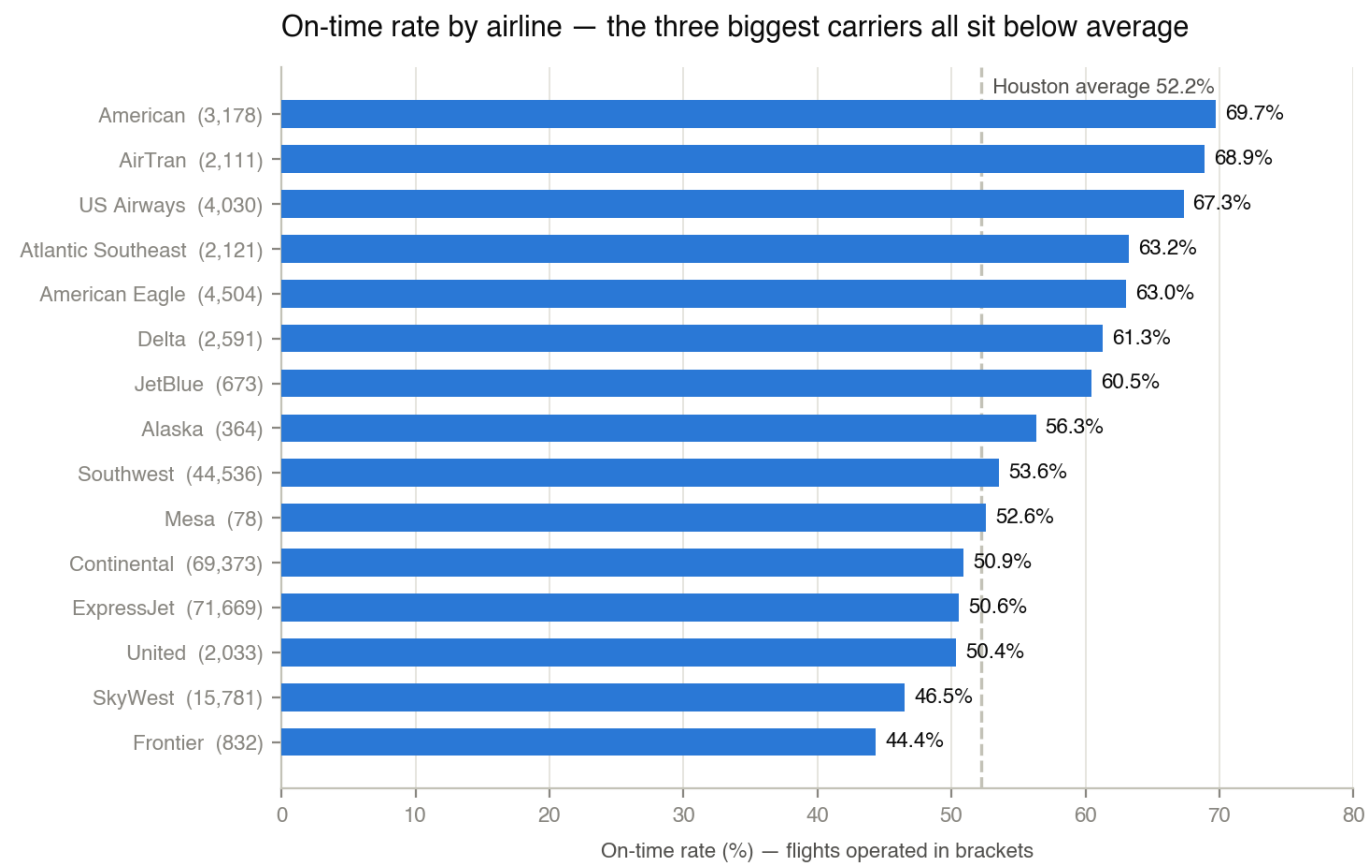
☐ Envoy Air

☐ ExpressJet

☐ Frontier Airlines

Part 6: Insights, Recommendations and Storytelling

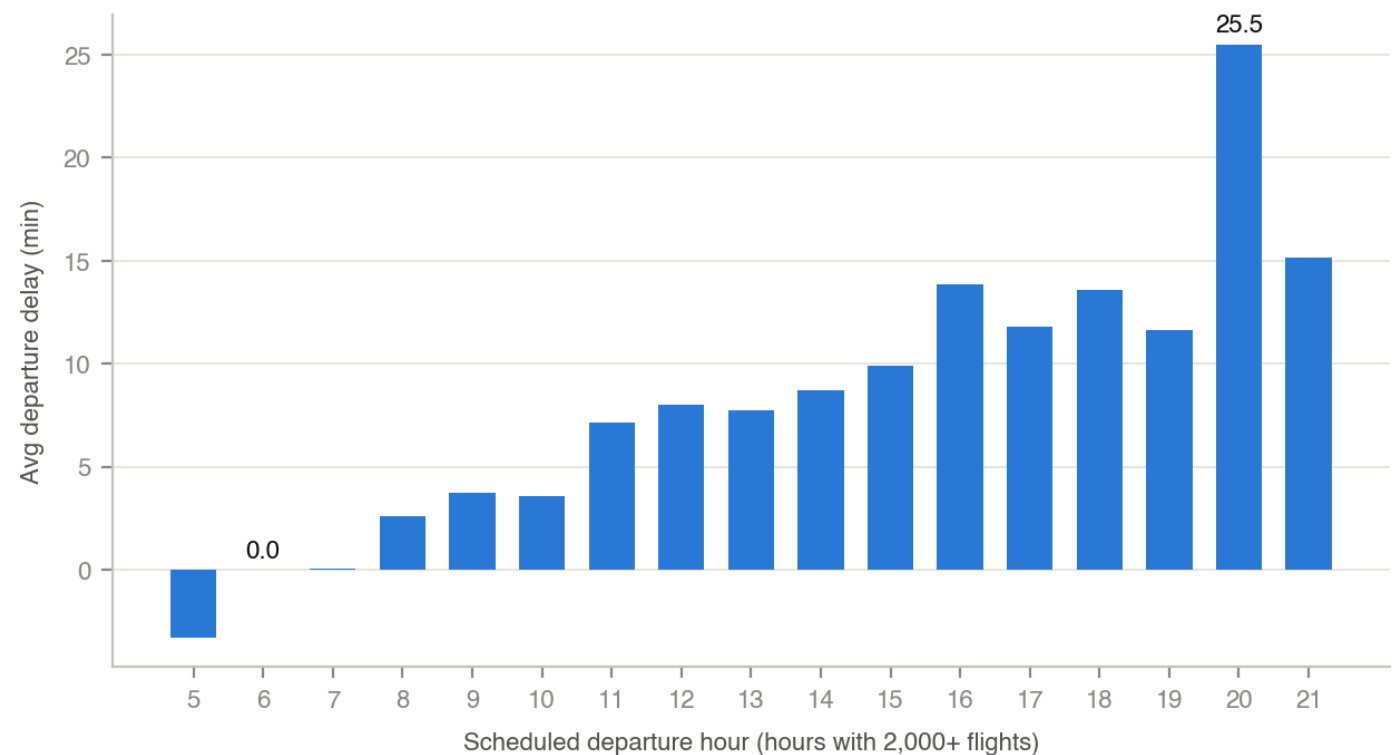
Insight 1 — Punctuality Is Inversely Related to Scale



Three carriers fly 82% of Houston. All three run at or below the 52.2% average. Improving the other twelve cannot move the system.

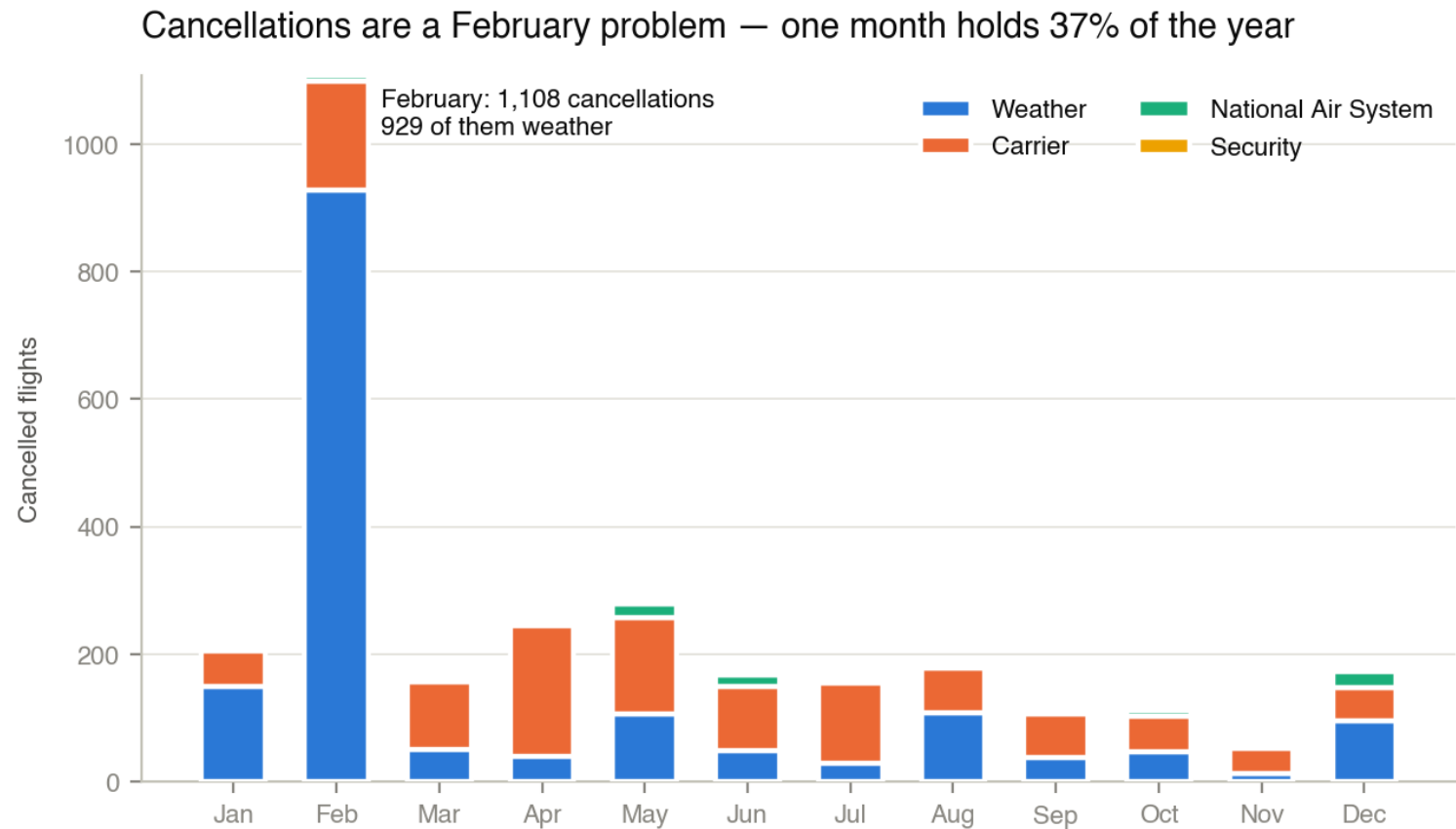
Insight 2 — Delay Is Manufactured During the Day

Delay compounds through the day — a 06:00 departure leaves on time, a 20:00 departure averages 25 minutes late



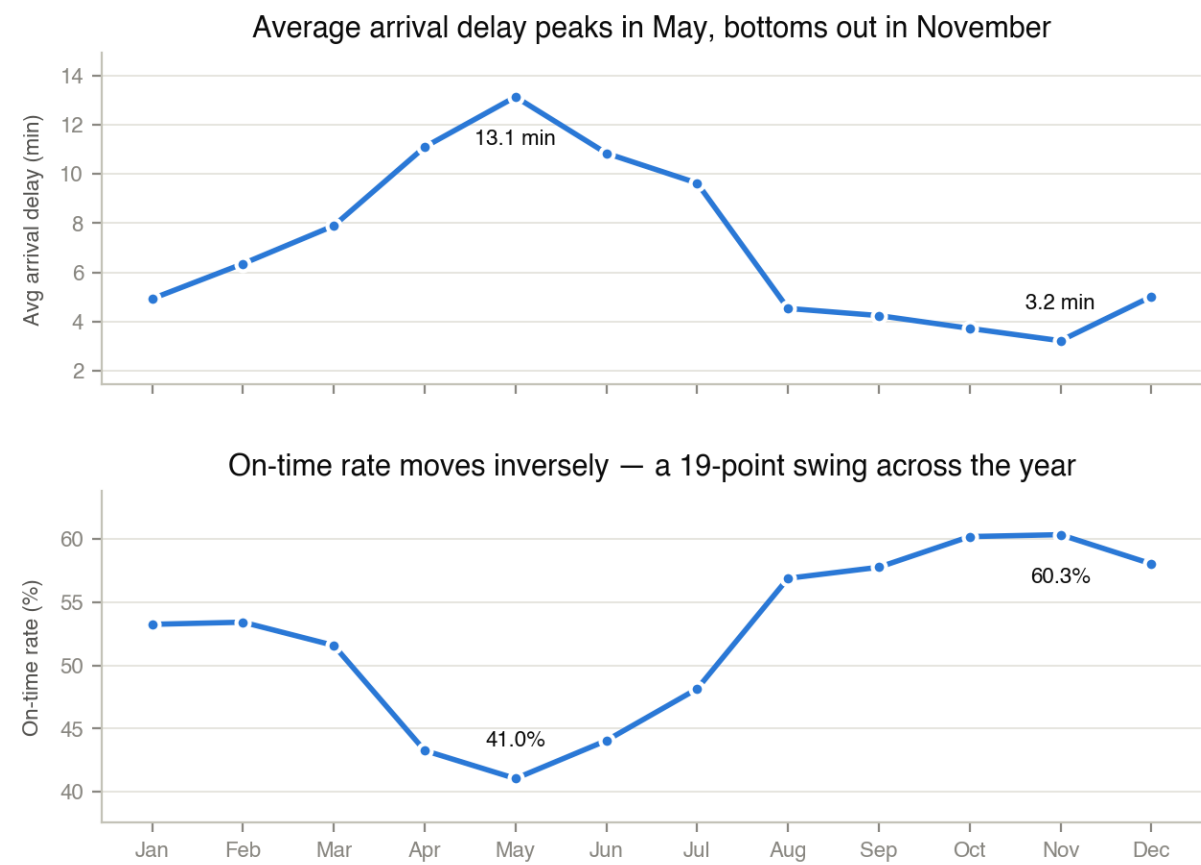
06:00 departs on time. 20:00 averages 25.5 minutes late. Same aircraft, same crews, same routes — delay compounds across rotations.

Insight 3 — Cancellations Are One Week in February



1,108 of 2,973 cancellations fall in February. Strip it out and the network cancels 0.89% — a well-run operation.

Insight 4 — Delay Peaks in May, Traffic Peaks in July



Volume does not predict delay: $r = 0.33$. Contingency staffing is aligned to the wrong month.

Insight 5 — A Thin Tail Does the Damage

Delay severity mix of 223,874 completed flights



On time / early Minor (1-14 min) Moderate (15-59 min) Severe (60-179 min) Extreme (180+ min)

79% of flights are effectively fine. The cost sits in the 4.7% — 10,584 flights — that arrive an hour or more late, of which 1,110 are over three hours late. An average-delay target chases the harmless middle.

What We Recommend

Six recommendations, traced to the evidence

#	Recommendation	Why
1	Target the three high-volume carriers	Moving them nine points moves the whole system seven. Perfecting the other twelve moves it under three.
2	Protect the morning bank	Delay compounds from 06:00. The earliest minute saved is the cheapest.
3	Fix HOU's turnaround, not its taxiways	Half the taxi time, more departure delay. The gate is the constraint.
4	Separate February from the cancellation baseline	The blended rate hides a good operation and misnames the dominant cause.
5	Target severe-delay rate, not average delay	Aims at the 10,584 flights that break connections, not at noise.
6	Load-plan for May, not July	Align contingency with when the network fails, not when it is busiest.

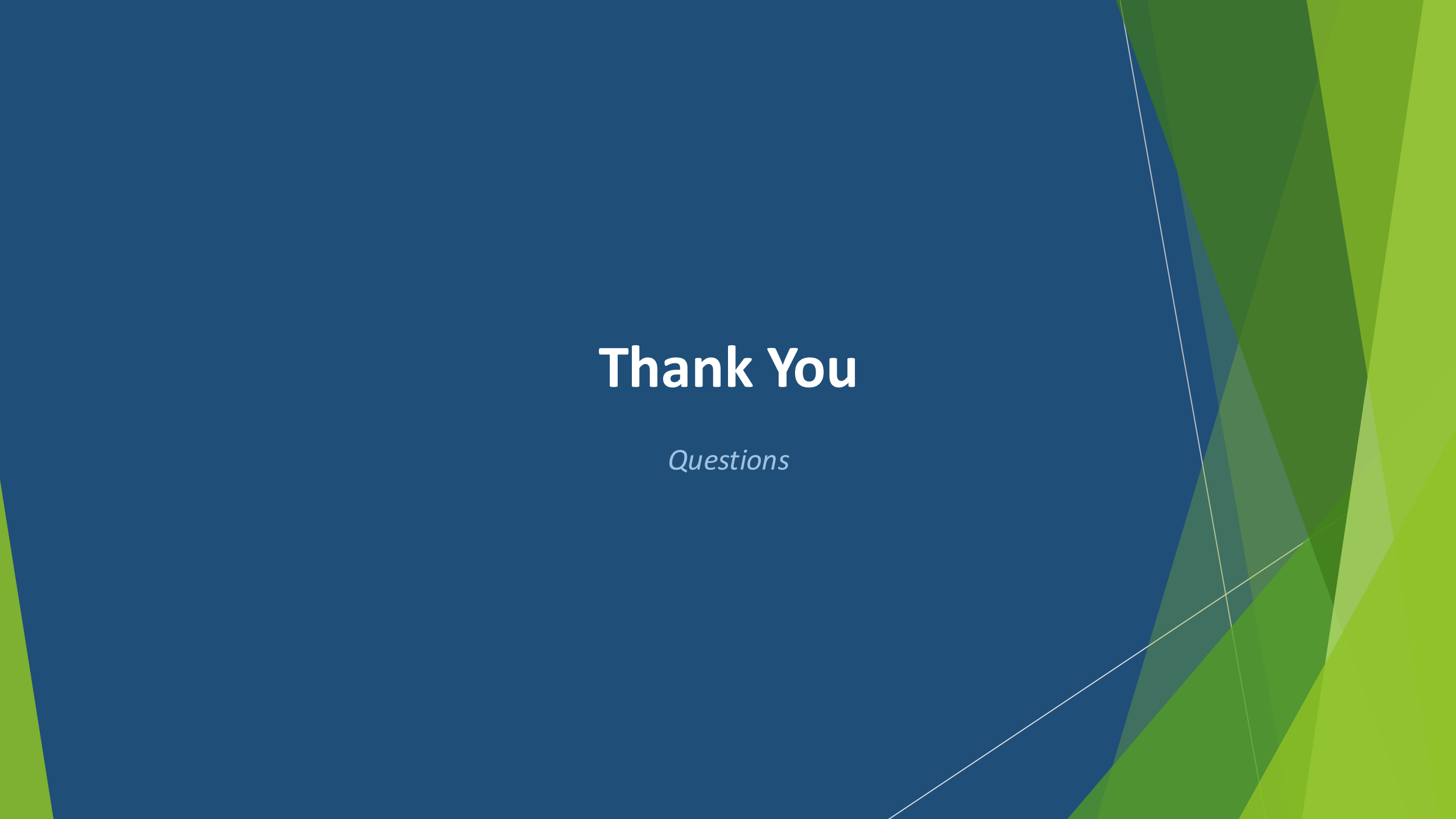
Where This Leaves Houston

- Houston does not have a general punctuality problem. It has three addressable problems and one unavoidable one.
- Unavoidable: February. One ice storm, 37% of the year's cancellations.
- Concentration: three carriers are the system.
- Compounding: delay is created hour by hour, not inherited.
- Measurement: the current metrics point management at the least actionable part of all three.

The cheapest recommendation — change what is measured — costs nothing and unlocks the value of the other five.

Recommendations for the Dashboard

- ① Investigate the specific days behind the three major delay spikes (around 500, 380, and 545 minutes) in Q1 2011 to identify whether they were caused by weather, mechanical issues, or operational disruptions, since these outliers have a bigger impact on passengers than the 6.86-minute average suggests.
- ② Set up daily delay alerts for any day where departure delay exceeds a set threshold (for example, 150 minutes), so severe-delay days like the ones seen in January and February 2011 are caught and addressed in real time rather than only in quarterly review.
- ③ Maintain the current zero-cancellation standard by documenting what operational practices kept cancellations at 0 in Q1 2011, and apply the same practices across other quarters and airlines.
- ④ Compare AirTran Airways' 6.86-minute average departure delay against the other airlines in the dataset (Alaska, American, Delta, Frontier, and others) to determine whether this performance is above or below industry average within the dataset.
- ⑤ Expand this analysis to a full-year, all-airline view before presenting to senior management, since the current figures are limited to one airline and one quarter and may not represent overall performance.

The background features a solid dark blue field on the left, transitioning into a series of overlapping, semi-transparent green and yellow-green geometric shapes on the right. These shapes are primarily triangles and polygons, creating a dynamic, layered effect. A thin white line runs diagonally across the lower right portion of the image.

Thank You

Questions